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ORIGINAL PAPER

Extensions of prescribed wake modelling for helicopter rotor BVI noise investigations

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Abstract The prediction of blade–vortex interaction noise is highly sensitive to the blade–vortex missdistance and thus requires the knowledge of the vortex position relative to the blade interacting with it. The generally accepted solution to the problem is to apply either free-wake codes (vortex lattice methods) or computational fluid dynamics (CFD) codes, both of which are very time consuming, especially CFD. Prescribed wake codes are computationally faster in comparison, since they are based on steady rotor operational conditions only, but they are considered to be not accurate enough due to the prescription of the geometry based on a few general parameters. First, they lack any wake geometry response due to variations in rotor loading distributions at constant thrust caused by, e.g., non-uniformity of the aerodynamic environment in forward flight, elastic blade motion, or by any means of active control. Second, wake deflections due to the presence of the fuselage are ignored. This paper provides an approximate solution to both problems, based on simple momentum theory considerations for the first problem, and based on the displacement flow of the fuselage for the second. This provides extensions for any existing prescribed wake model to account for rotor loading distribution and fuselage effects on the prescribed wake geometry to a first order of accuracy, sufficient for the investigations of the sensitivity analysis of noise radiation due to variations of blade design or due to applications of active control.

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Keywords Prescribed wake · Wake distortion · Active rotor control · BVI · Rotor noise · Rotor wake simulation · Harmonic loading · Fuselage/wake interference

Abbreviations

B	Tip loss factor = effective non-dimensional blade radius
c_{nj}	Fourier coefficient of $q_n(\psi)$
C_T	Thrust coefficient, $C_\mathrm{T} = T/(\rho\pi\Omega^2 R^4)$
$f(r)$	Radial distribution function
f_m	Weight factor of $f(r)$
k_x, k_y	Longitudinal and lateral inflow gradients
L_n	Blade lift at the nth harmonic, N
$L_{n\mathrm{S}}$, $L_{n\mathrm{C}}$	Sin and cos part of L_n
n, m	Integer number
N_b	Number of blades
N_rev	Number of wake revolutions
N_R	Number of blade elements
N_ψ	Number of vortex segments per revolution
$q_n(\psi)$	Fourier series of the nth mode of the fuselage-induced flow
r	Non-dimensional radial coordinate
r_a	Non-dimensional root cut-out radius
r_vr	Radius of vortex release point
R	Rotor radius, m
t	Time, s
t_vr	Time of vortex release, s
T	Thrust, N
v_i	Induced velocity normal to the disk, m/s
v_if	Fuselage-induced velocity, m/s
V	Velocity, m/s
V_∞	Flight speed, m/s
x, y, z	Non-dimensional coordinates
x_e	Rear end of the disk
x_v, y_v, z_v	Vortex position

x_{vr}, y_{vr}, z_{vr}	Coordinates of vortex release point
z_{te}	Vertical position of trailing edge
α	Tip path plane angle of attack, rad
λ_i	Induced inflow ratio, $\lambda_i = v_i/(\Omega R)$
λ_{i0}	Mean-induced inflow ratio
λ_{in}	Induced inflow ratio, nth harmonic
λ_{inh}	Induced inflow ratio computed in hover
μ	Advance ratio, $\mu = V_\infty \cos\alpha/(\Omega R)$
μ_z	Axial inflow, $\mu_z = -\mu \tan\alpha$
ρ	Air density, kg/m^3
ϕ_n	Radial distribution function
ψ	Azimuth, $\psi = \Omega t$, rad
ψ_b	Blade azimuth at vortex release, rad
Ω	Rotor rotational frequency, rad/s

1 Introduction

The prediction of rotor noise radiation is important for helicopter certification and operation [1]. Especially in descending flight, blade–vortex interaction (BVI) noise is generated by the main rotor blades in the first and fourth quadrant of the rotor disk. There, the blade leading edge and the blade tip vortices of preceding blades (or of the blade of investigation) are almost parallel to each other, causing a high frequency surface pressure variation simultaneously over a large part of the rotor blade. The major parameter of noise radiation—besides the vortex circulation strength, blade Mach number and blade loading during the interaction—is the vertical blade–vortex miss-distance [2]. Thus, the prediction of this missdistance is of utmost importance when computing BVI noise. While the computation of blade air loads and the blade dynamic response associated with these is state of the art and can be reliably performed with comprehensive codes, the prediction of the wake geometry is still a challenging task and was not possible at all until about 1960.

In the 1960s, computers became available and made possible the application of computationally intensive methods to engineering problems. Since then, computer power has quickly developed and with it the possibility to apply more and more complex algorithms, opening the door for BVI noise prediction using different levels of sophistication. A good overview of available engineering methods of rotor wake prediction and their application is given in [3]. Essentially, three methods are applied today for computation of BVI noise radiation.

First are prescribed wake codes, where the wake geometry is purely defined by a few parameters, namely the thrust coefficient, the advance ratio, the climb or descent velocity, and the tip path plane angle of attack. Based on the experimental data in hover obtained in 1956 [4], the first simple prescribed wake geometry model was developed. More recognized is the experimental work and math modeling of Landgrebe [5] in 1971 and later. Once the wake geometry is computed, it is held constant, independent of how the azimuthal loading distribution of a rotor blade may vary during its revolution. The major advantage is that only the induced velocities of all vortices of the wake at the aerodynamic point of interest of the revolving blade need to be computed, which is associated with very moderate computational amount. Probably, the latest development was given by Beddoes [6] that included a simple approximation to the lateral roll-up of the wake system. The major shortcomings of classical prescribed wake codes are that they include no response of the wake geometry to load variations within the disk (for example, as caused by means of higher harmonic or individual blade control) for otherwise constant steady operating conditions of the rotor. These load variations have been demonstrated in wind tunnel and flight tests to be the major source of noise reduction.

Second are free-wake codes, which can be represented by a vortex lattice method. Therein, at any instant of time the differential equations of motion of every individual vortex element within the entire wake is solved. Thus, the shortcomings of classical prescribed wake codes are not present in free-wake codes. This has resulted in the rotorcraft community assuming that the usage of free-wake codes is mandatory when active rotor control is to be investigated, especially for BVI noise radiation. This requires the computation of induced velocities of all vortices in the wake on all lattice nodes. This is a tremendous effort, considering a representative wake of a four-bladed rotor consists of about $N_b N_\psi N_{rev} = 4 \times 72 \times 5 = 1440$ vortex segments only for the tip vortices, even with truncation after the fifth wake revolution. With all inboard wake vortex filaments, this number is considerably higher since this includes $N_R + 1$ trailed vortices and $N_R N_\psi$ shed vortices, both per blade. It is immediately apparent that this is associated with a tremendous increase of computing effort compared to the prescribed wake approach, but the method is inherently capable of responding to any kind of blade loading distribution. Consequently, the resulting wake geometry can be expected to be much more accurate, and this is the reason why the majority of BVI noise prediction attempts are based on free-wake codes. The application of such codes started by Scully [7] and was increasingly used and developed since then by numerous authors; the latest developments have been generated by Leishman et al. [8].

The third method is the use of computational fluid dynamics (CFD) codes, which attempt to solve the fluid differential equations of motion (Navier–Stokes or Euler formulation) in space and time. This is performed by discretizing the entire volume of space around the rotor into millions of cells and solving these equations numerically

for each of these cells. It is associated with a tremendous increase of computational effort compared to free-wake codes. Additional complications arise from the need for a fine discretization around vortical structures and the fact that the location of these is a function of the solution itself. Inherent numerical dissipation of CFD codes is a further problem, and the application of CFD for BVI noise prediction is still in its infancy today. Yet CFD results have the potential of being superior to free-wake results with respect to accuracy. An example of using CFD for rotor BVI noise application is given in [9].

Considering the computational effort required by prescribed and free-wake methods, the following estimate can be calculated. In a prescribed wake, only the induced velocities on the blades have to be computed using the Biot–Savart law. For any individual collocation point on one of the blades at a given azimuth, the fully discretized wake consists of $N_b(N_R + 1)(N_\psi\ N_{rev})$ trailed and $N_b N_R N_\psi\ N_{rev}$ shed vortex elements. The total number of Biot–Savart law evaluations thus is

$$N_{presc} = N_b(2N_R + 1)(N_\psi N_{rev}) \tag{1}$$

for every individual collocation point anywhere on the rotor disk. Next, consider a free-wake model that applies the same discretization. Additional to the induced velocities on the blades (N_{presc}) these have to be computed on all nodes of the wake, i.e. on $N_b(N_R + 1)N_\psi\ N_{rev}$ nodes, for each time step for every vortex element that is present in the azimuthal direction $N_\psi\ N_{rev}$ in order to obtain a converged periodic solution for the wake geometry. Therefore,

$$N_{free} = N_{presc}\left\{1 + [N_b(N_R + 1)N_\psi N_{rev}]N_\psi N_{rev}\right\} \tag{2}$$

The ratio of computational effort of prescribed wake to free-wake then is

$$\frac{N_{presc}}{N_{free}} = \frac{1}{1 + [N_b(N_R + 1)N_\psi N_{rev}]N_\psi N_{rev}} \approx \frac{1}{N_b N_R (N_\psi N_{rev})^2} \tag{3}$$

As an example, a four-bladed rotor $N_b = 4$ may be discretized into $N_R = 20$ blade elements (this is the lower end for acoustic investigations; a fine resolution would have 40–50 elements). If the wake elements may have a length of 10° azimuth (about the coarsest resolution, i.e., $N_\psi = 36$, a fine resolution would have elements no longer than 5°, i.e. 72 per revolution) and the wake may extend over only $N_{rev} = 2$ revolutions in length (this covers just the rotor disk as minimum requirement; a usual length would be 4–5 revolutions). Then, the above ratio amounts approximately to $1{:}4 \times 10^5$. If only the tip vortex is retained and all other trailed and shed vortices are ignored (with the reasoning that the tip vortex strength is about 10 times larger than all others), then this reduces to approximately $1{:}2 \times 10^4$. Including all kinds of acceleration measures (for example: computing the wake–wake interaction only for vortex elements within a certain distance to the node of interest), this can be further reduced to about $1{:}10^3$, while these numbers are based on the minimal discretization.

As a rough estimate, the relation of computational effort can thus be assumed very conservatively as $1{:}10^3{:}10^6$ for prescribed wake:free-wake:CFD. Therefore, prescribed and free-wake codes are favourable for routine application and the free-wake approach was seen as the best compromise of computational effort and accuracy obtainable. However, it appears that the potential advantages of prescribed wake codes are not fully exploited today. As will be shown in this paper, the wake geometry modifications due to azimuthal loading variation of the rotor blades, as well as the effect of fuselage displacement flow on the wake geometry can be accounted for in a prescribed wake to a first order of accuracy. This extends the usage of prescribed wakes significantly, especially when a large number of parameter variations is to be investigated with respect to its effect of BVI noise radiation.

An example of a free-wake result is given in Fig. 1. Although the fuselage is shown below the rotor, its effect on the wake distortion was neglected and the results are based on an isolated rotor. Yet, some fundamental general behaviour can already be identified qualitatively:

- In the front of the disk, the vortices are convected upwards above the disk. Towards the rear and behind, a strong downwash is present in the middle of the first and fourth quadrant, indicating a strong longitudinal gradient of the induced velocities.
- In the middle of the disk and behind it, in a lateral extension of about the root cut-out area, the vortices remain above the disk before being convected downwards further behind the disk.
- On the right and left sides of the disk, the tip vortices bundle around each other and remain above the disk as if an upwash is present there. This is the roll-up region, and far behind the rotor this appears very similar to the wake of a fixed wing aircraft with two long straight tip vortices.

Within a comprehensive rotor simulation environment, all of these effects must be addressed appropriately to at least a first order of accuracy. In addition, the proximity of the fuselage to the rotor disk should also be taken into account, which currently is not done.

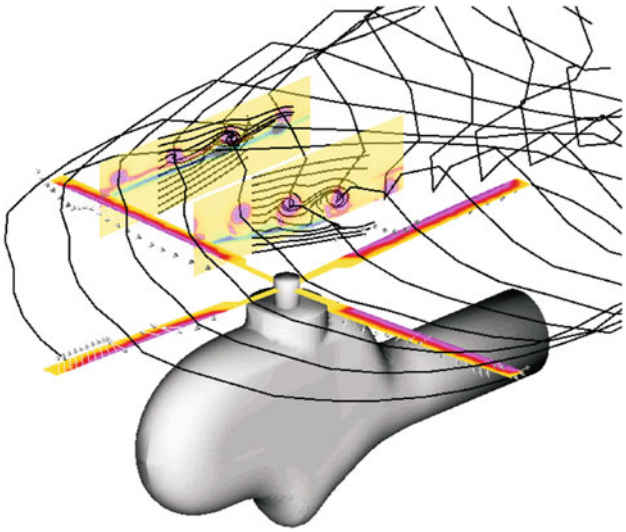


Fig. 1 Free-wake result for the wake geometry in descent

2 Fundamentals of prescribed wake computation

The prediction of the wake geometry requires the knowledge of the flow field wherein the vortex elements are convected downstream (mainly due to flight speed), in vertical direction (mainly due to self-induced velocities, but also due to fuselage flow displacement) and in lateral direction (to a lesser extent, due to the same reasons). The only known parameters are the flight velocities in all directions and the rotor disk attitude. However, the latter already implies some inaccuracy since the rotor blades are flexible, and the actual shape of the tip path plane is deformed rather than planar. Mostly unknown are the self-induced velocities, which in reality are extremely complex and non-linear with respect to both the radial and azimuthal directions. Therefore, several simplifying assumptions must be established in order to allow a wake geometry to be computed analytically, based on a few available parameters.

First, the induced velocities are assumed to be independent of the vertical coordinate. For the moderate vertical deflections of the wake geometry this is considered not to be an acceptable simplification since the self-induced velocities originate from the wake itself. Thus, whatever deflection the wake undergoes, the relative position of one vortex element to another does not show large variation.

Second, the non-uniformity of the induced velocities may be replaced by a simple linear variation in the longitudinal and lateral direction based on elementary momentum theory considerations. This is a strong simplification and thus most questionable. However, high frequency variations of induced velocities (which is equivalent to local spatial variations) that may be present along the convection path of a vortex element will be passed quickly and thus does not cause large wake displacements. Of more concern are low frequency velocity field oscillations (or, in spatial terms: long wave length distributions) since they affect the vortex path for a longer time and will cause larger and more significant deflections.

Therefore, it is important to consider this low frequency content of induced velocities when computing the prescribed wake geometry.

2.1 Rotor-induced velocity field

The most simple assumption regarding the induced velocity distribution within the rotor disk is to apply momentum theory to the rotor thrust for the operational condition under consideration. The fundamentals of momentum theory were developed by Rankine [10] and Froude [11], and the application to rotary-wing problems first was performed by Glauert [12], then by Lock [13]. Glauert already suggested a linear longitudinal variation of induced velocities, which was then applied by Lock. A lateral gradient was introduced by Drees [14]. Thus, a simple induced velocity field within the rotor disk may be written in the form

$$\lambda_i(x,y) = \frac{v_i(x,y)}{\Omega R} = \lambda_{i0}\left(1 + k_x x + k_y y\right) \tag{4}$$

with $x = r\cos\psi$ and $y = r\sin\psi$. The definition of the geometry is given in Fig. 2. The gradients of this variation were assumed by Glauert and Lock to be in the range of $k_x \approx 1$ and are dependent on flight velocity (since in hover they have to be zero). The computation of these gradients was attempted much later by several authors, for example, see [15] and [16]. The results for k_x are highly dependent on the method applied, and even can differ by 100% for identical operational conditions. Only Drees provided a value for the lateral gradient, $k_y = -2\mu$. The approximation of a highly non-linear distribution by only a linear gradient (or, a 1/rev variation for the rotor blade) must be seen as a crude estimate, but has proved sufficient for rotor performance investigations. To overcome this problem it has been demonstrated in [22] that an iterative method is useful. Here the gradients are computed from the distribution of the vortex induced velocities of the prescribed wake within the rotor disk. In the first step, the gradients are taken from any of the available theories, and in all further iterations they are taken from the Fourier analysis of the resulting induced velocity field. The practical application in [22] has shown that these gradients quickly converge.

In forward flight, the mean value of induced velocities, λ_{i0}, is a non-linear function of the hover-induced velocity $\lambda_h = \sqrt{C_T/2}$ (and thus a function of the thrust coefficient), the advance ratio μ, and the axial inflow μ_z due to the tip

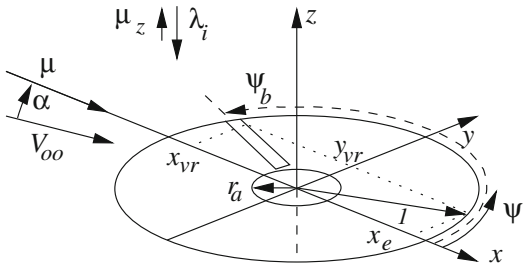


Fig. 2 Definition of the geometry

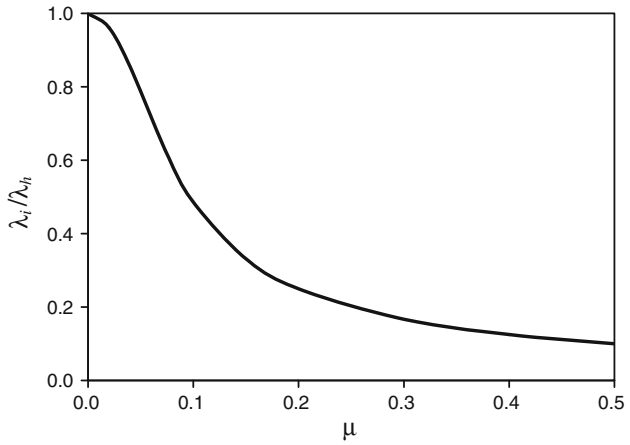


Fig. 3 Mean induced velocity, $\alpha = 0$

path plane angle of attack. In the general form—as given by Eq. 5—it must be solved iteratively.

$$\lambda_{i0}(\mu, C_T) = \frac{\lambda_h^2}{\sqrt{\mu^2 + (\mu_z + \lambda_{i0})^2}} \tag{5}$$

This is a result of application of momentum theory to forward flight, see [15, 16], and has no general analytical solution for the induced inflow, except for some special cases, e.g., $\alpha = 0$. In hover, this immediately equates to $\lambda_{i0} = \lambda_h$ as required, and in high speed forward flight, where $\mu \gg \mu_z + \lambda_{i0}$, it equals to $\lambda_{i0} \approx \lambda_h^2/\mu$. For the special case when $\alpha = 0$, the solution is shown in Fig. 3. In hover, the total momentum to generate thrust is by means of accelerating the flow, while in forward flight a large portion of the momentum is already provided by the speed of flight, and only little additional acceleration of the flow is needed to generate the thrust. A typical landing approach, where BVI noise is dominant is performed at an advance ratio of $\mu \approx 0.15$, where already a significant reduction of the mean-induced flow is present (30% of the induced velocity in hover).

A higher-order model for the induced velocities within the rotor disk was given by Mangler and Squire [17]. Shape functions were used for the radial distribution and a Fourier series for the azimuthal variation. The amplitudes depend on the tip path plane angle of attack and the order of harmonic, which consist of even multiples of the rotor revolution $n = 0, 2, 4, \ldots$ and only the $n = 1$ and 3 of the uneven harmonics. The shape functions are highly nonlinear and depend on the order of harmonic of the Fourier series, where only cos-terms appear due to the lateral symmetry. With $s_r = \sqrt{1 - r^2}$ and $r^2 = x^2 + y^2$ these shape functions read

$$f_0 = r^2 s_r \quad f_1 = r(9r^2 - 4) \quad f_3 = r^3$$
$$f_n = \left[3s_r + (n + s_r)\frac{n^2 + 3 - 9r^2}{n^2 - 1}\right]\left(\frac{s_r - 1}{s_r + 1}\right)^{\frac{n}{2}} \tag{6}$$
$$n = 2, 4, 6, 8, \ldots$$

2.2 Approximation of roll-up

So far, steady and 1/rev variations of induced velocities have been addressed. Additionally, it is known from experiments that a lateral roll-up is also present on both sides of the rotor disk, which is not taken into account so far. Beddoes [6] suggested a lateral modification of the global inflow proportional to $-k_x|y|^3$ in order to obtain a wake geometry closer to measurements in these areas. Due to the absolute value of y, this results in a violation of the mean inflow and thus the momentum theory result, which was corrected in [18] by replacing the 1 in Eq. 4 by the factor k_0. Then, the induced velocity field is given by

$$\lambda_i(x, y) = \lambda_{i0}\left[k_0 + k_x\left(x - |y|^3\right) + k_y y\right]$$
$$\text{with } k_0 = 1 + \frac{8k_x}{15\pi} \tag{7}$$

For $k_x = 1$, the correction factor in k_0 amounts to $8/(15\pi) = 0.17$, an important correction that should not be ignored.

2.3 The influence of tip loss and root cut-out

Virtually, all helicopter rotors include a root cut-out with the airfoil-shaped section of the rotor blade beginning at about $r_a = 0.2$. At the blade tip the lift vanishes, so that an effective radius can be assumed [15, 16]. With respect to momentum theory, the tip loss is accounted by a tip loss factor B to define an effective rotor radius. It is a function of number of blades and thrust coefficient; for conventional rotors it is of the order of $B = 0.98$. This is usually ignored, but can be accounted for in a simple manner. In this case, the momentum theory is applied only to the lifting part of the rotor disk, which now has a disk area reduced by the tip loss area, i.e., $\pi(1^2 - B^2) \approx 0.04\pi$ and the root cut-out area, i.e., $\pi(r_a^2 - 0^2) \approx 0.04\pi$. Both effects are the same order and reduce the total disk area by 4% each. A constant mean-induced velocity distribution must therefore be corrected by the factor

$$\frac{1}{\sqrt{B^2 - r_a^2}} \approx 1.04 \tag{8}$$

and can be included into the factor k_0 in Eq. 7. This may be seen as a minor correction, but it is sensitive to the effective rotor radius and root cut-out. In case of $B = 0.96$ and $r_a = 0.3$ (= a 17% change in the effective disk area), this correction already amounts to factor of 1.1. When the Mangler model is used, even the steady distribution is a function of the radial coordinate, with zero-induced velocity located at the center of the disk and at the blade tip. In this Mangler model, the correction can be ignored since the main momentum has already shifted to the airfoiled section of the blade and the reduced area of the disk (due to B and the root cut-out) generates only marginal corrections as shown in [18].

Thus, when considering the root cut-out, the mean velocity of the remaining part of the disk is increased, and inside the root cut-out area, the induced flow should be set to zero: $\lambda_i(r < r_a) = 0$. This is important to correctly model the vertical wake convection through this inner area of the disk. The assumption of zero-induced flow in the root cut-out area is a consequence of the actuator disk theory, and in a real situation (or as evinced by CFD calculations) there is indeed a remaining induced flow in this region. To account for such more accurate refinements is left for future investigations.

2.4 Induced velocity field behind the rotor

For computing the wake geometry, not only the velocity field inside the rotor disk is needed but also the velocity field behind it. Since the further deformations behind the disk are of no importance for the BVI problem inside the disk, more rigorous assumptions as to the distribution of induced velocities are allowed. The most simple one is to assume a constant mean induced velocity everywhere without any further gradients. For the longitudinal gradient, this is justified since at any location behind the disk the part of wake generating significant influence on a vortex filament is as long ahead of it as behind of it. This symmetry precludes a gradient of induced velocity in x direction. A lateral gradient may still exist due to different vortex circulation strengths on the advancing and retreating side, following the result of Drees [14]. The same applies for the roll-up due to Beddoes [6]; this process is initiated within the disk and continues to develop in the wake behind.

Another way of examining the problem may again be taken from momentum theory. Far from the rotor (in hover: underneath it; in forward flight: behind it), the induced slipstream is fully developed and has twice the speed it had within the rotor, i.e. $\lambda_i = 2\lambda_{i0}$. In order to ease the analysis, this result may be evaluated at the rear of the rotor disk and applied everywhere downstream. As a result, the induced velocity field in the wake behind the rotor disk is

$$\lambda_i(x, y) = 2\lambda_{i0}\left(k_0 - k_x|y|^3 + k_y y\right) \tag{9}$$

with k_0 as defined in Eq. 7. As an example, in Fig. 4 the induced velocity distributions within the rotor disk of the four models of Glauert, Lock, Drees and Beddoes are given. For convenience, the induced flow has been referenced to the mean value, and the scale is denoted by λ_n/λ_0, which in this case is $\lambda_i(x, y)/\lambda_{i0}$.

2.5 Operating condition

The reference operating condition used throughout the paper is the HART II baseline (BL) case [24]. The HART II wind tunnel test was performed in 2001 in the 8 m × 6 m open-jet configuration of the large low-speed facility of the DNW by an international team of DLR, Onera, NASA Langley, US Army AFDD and DNW. A 40% Mach scaled hingeless Bo105 model rotor was used to explore the effect of 3/rev HHC on the wake geometry and the noise radiation. A rotor blade was equipped with absolute pressure sensors along the leading edge for detection of BVI locations and at one radial position for analysis of the section airloads. The BL case is defined by an advance ratio of $\mu = 0.15$, a thrust coefficient of $C_T = 0.0044$ and a tip path plane angle of attack of $\alpha = 4.5°$, including wind tunnel corrections. This represents a descending flight where strong BVI occurs, i.e., BVI takes place around $\psi = 50°$ and $\psi = 300°$ within the rotor disk. There, the vortex axis and the blade leading edge are approximately parallel to each other. In addition to the BL case, two cases with 3/rev HHC were investigated: one with a 3/rev control phase of 300° that was found to generate minimum noise radiation (MN) and another one with a control phase of 180° for minimum vibration generation (MV). A large portion of the HART II data including all documentation was made available for the HART II International Workshop and is accessible for download from NASA (http://rotorcraft.arc.nasa.gov/hart2/) or DLR (ftp://HART-II@ftp.dlr.de/). The data have been widely used in the community for code validation.

2.6 Computation of the wake geometry

Any vortex element of the wake was released at the trailing edge of its generating blade. In most cases this will be at the blade tip, but, depending on the radial distribution of blade loading, this may also be at a more inboard location. In cases where the blade tip region experiences a download and the inner part of the blade an upload two vortices will be released: one at the tip with opposite sense of rotation

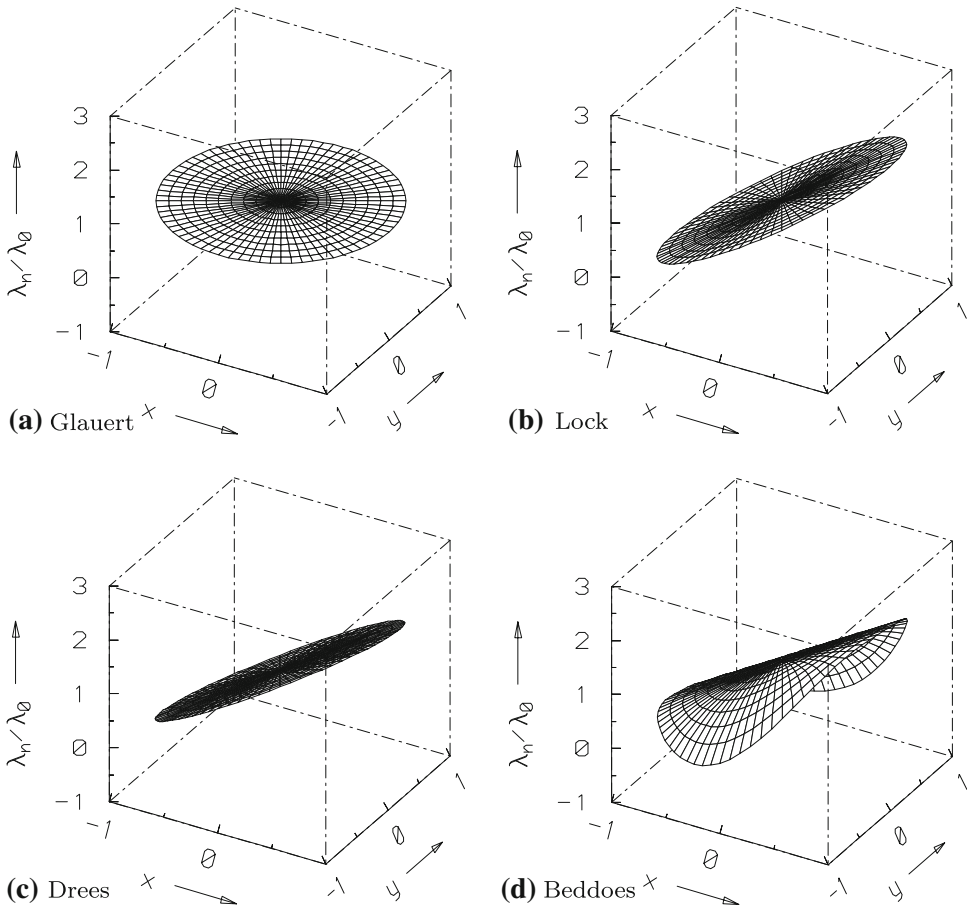


Fig. 4 Distribution of induced velocity for the various models

due to the download and another one inboard where the radial gradient of bound circulation is maximum. This inboard vortex will have the conventional sense of rotation. Since a download in the tip region usually is present only over a part of the revolution such dual vortex systems are developing only in these regions and everywhere else a single tip vortex is existing.

For every azimuthal position of the blade, the tip vortex release point (and that of the secondary vortex, if present) is defined by the azimuth of the generating blade ψ_b, the radial release position r_{vr}, and the trailing edge position vertically with respect the hub plane, z_{te}, which is given by the blade flapping position of this radius, and by the total local blade pitch due to the blade pre-twist, the control angle (including active control, if applied) and the elastic torsion. Thus, the coordinates of the vortex release point are

$$x_{vr} = r_{vr} \cos \psi_b; \quad y_{vr} = r_{vr} \sin \psi_b; \quad z_{vr} = z_{te} \tag{10}$$

From this point, the vortex convects downstream along x by the in-plane component of the oncoming velocity $V_\infty \cos \alpha$ or, in non-dimensional form: the advance ratio μ. It convects vertically depending on the distribution of vertical velocities along its downstream convection, that are the induced velocities v_i and the out-of-plane component of the oncoming velocity $-V_\infty \sin \alpha$ or, in non-dimensional form: λ_i and μ_z. Since disturbances of the flow field are mainly in the vertical direction and less in the downstream and lateral directions, the focus here is placed only on the vertical convection and associated vortex trajectories. In general, disturbances in any direction could be handled the same way, but this is left for future investigations. Any vortices generated at the front of the disk, i.e., between $\pi/2 < \psi_b < 3\pi/2$, have to pass through the rotor disk and the induced velocity field therein, while all elements generated at the rear of the disk, i.e., between $3\pi/2 < \psi_b < 2\pi$ and $0 < \psi_b < \pi/2$, are always behind the rotor disk and are only affected by the induced velocities there.

The vertical velocity field through which a vortex element has to pass consists of two contributions. First, resulting from the inclination of the tip path plane with respect to the flight direction, $\mu_z = -\mu \tan \alpha$ is constant across and behind the disk. Second, the induced velocity $\lambda_i(x, y)$ acts normal to the tip path plane, which, since the inclination with respect to the hub plane is usually small, can be assumed to be normal to the hub plane. Thus, the total velocity normal to the hub plane is given by $\lambda(x, y) = \lambda_i(x, y) + \mu_z$.

When using the Mangler model [17] (which is expressed in polar coordinates, i.e., by r and ψ), a coordinate transform is required since the downstream convection of the vortex occurs along x. Thus, it is needed to express the induced velocities as a function of x and y instead of r and ψ. For the first harmonic this is an easy task ($\cos\psi = x/r$, $\sin\psi = y/r$), but the higher harmonics require the application of Moivre's formula [19] to express them as powers of the first harmonic. For example, the term of the 6th harmonic and the associated amplitude have the following expressions ($s_r = \sqrt{1-r^2}$, $r^2 = x^2 + y^2$), see Eq. 6:

$$\begin{aligned}\cos 6\psi &= 1 - 15\sin^2\psi + 45\sin^4\psi - 32\sin^6\psi \\ &= 1 - 15\frac{y^2}{r^2} + 45\frac{y^4}{r^4} - 32\frac{y^6}{r^6} \\ f_6 &= \left[3s_r + (6+s_r)\frac{39-9r^2}{35}\right]\left(\frac{s_r-1}{s_r+1}\right)^3\end{aligned} \tag{11}$$

Thus, for this harmonic (and the lower ones, too), the expressions of induced velocity within the disk are rather complicated which makes them difficult—but not impossible—to integrate analytically along x.

For simplicity, the remaining part of the paper makes use of the formulation of the induced velocity field following Drees [14] and Beddoes [6], but there is no compelling reason not to use the more complicated Mangler model other than the mathematical difficulties encountered. Whatever induced velocity model may be used, the vertical position of a vortex filament at any time t after its release from the blade at t_{vr} traveling downstream across the rotor disk is defined by the distribution of induced velocities it has to pass along its way.

$$z_v(x,y) = -\frac{1}{R}\int_{t_{vr}}^{t} v_i(x,y) - V_\infty \sin\alpha \,\mathrm{d}t \tag{12}$$

With $t = x(\Omega R)/(\Omega V) = x/(\Omega\mu)$ for small tip path plane angles usually encountered, and as well $\mathrm{d}t = \mathrm{d}x/(\Omega\mu)$. Thus,

$$z_v(x,y) = -\frac{1}{\mu}\int_{x_{vr}}^{x} \lambda_i(x,y) + \mu_z \,\mathrm{d}x \tag{13}$$

Within the disk two different regions may be encountered. Within $r < r_a$ the vortex element is passing the root cut-out area, where no induced velocities are present (see comment in Sect. 2.2). This affects all downstream positions if they are located in the lateral region $-r_a < y < r_a$. In all other areas of the disk, the vortices continuously travel within the induced velocity field until leaving at the rear side.

First, considering the latter case, where an element has traveled from the front of the disk (i.e., $x_{vr} < 0$) to any point $x \le \sqrt{1-y^2}$ still inside the disk experiences:

$$\begin{aligned}z_v &= -\frac{1}{\mu}\int_{x_{vr}}^{x} \lambda_{i0}\left[k_0 + k_x\left(x - |y|^3\right) + k_y y\right] + \mu_z \,\mathrm{d}x \\ &= -\frac{\lambda_{i0}}{\mu}\left[\left(k_0 - k_x|y|^3 + k_y y\right)(x - x_{vr}) + \frac{k_x}{2}\left(x^2 - x_{vr}^2\right)\right] \\ &\quad + \tan\alpha(x - x_{vr})\end{aligned} \tag{14}$$

where $y = y_{vr}$ and $\tan\alpha = -\mu_z/\mu$, assuming no further longitudinal and lateral velocity disturbances are present. This integral is valid until the element reaches the end of the disk at $x_e = \sqrt{1-y_{vr}^2}$. Behind the disk, for $x > x_e$, the convection is defined by a more simple model of induced velocity, $\lambda_i = 2\lambda_{i0}$ as outlined before and thus, with the induced flow from Eq. 9,

$$\begin{aligned}z_v &= z_v(x_e) - \frac{1}{\mu}\int_{x_e}^{x} 2\lambda_i + \mu_z \,\mathrm{d}x \\ &= z_v(x_e) - \frac{2\lambda_{i0}}{\mu}\left(k_0 - k_x|y|^3 + k_y y\right)(x - x_e) \\ &\quad + \tan\alpha(x - x_e)\end{aligned} \tag{15}$$

In the second case, for vortices generated at the front of the disk between $-r_a < y < r_a$, Eq. 14 is used until $x = -\sqrt{r_a^2 - y^2}$. Then, for the range $-\sqrt{r_a^2 - y^2} < x < \sqrt{r_a^2 - y^2}$, no induced velocity is acting, and the only remaining vertical velocity is due to flight speed and α. For this inner part of the disk

$$z_v = z_v\left(x = -\sqrt{r_a^2 - y^2}\right) + \tan\alpha\left(x + \sqrt{r_a^2 - y^2}\right) \tag{16}$$

After the element has exited the root cut-out area at $x = \sqrt{r_a^2 - y^2}$, Eq. 14 is applied again, but with $\sqrt{r_a^2 - y^2}$ replacing x_{vr}. The general effect of this root cut-out is to raise the wake in descent within and downstream of the root cut-out area. In climb, the vertical velocity due to flight speed and α is opposite to that in descent and hence the effect is vice versa. The maximum effect occurs at $y = 0$ the centreline.

The third case applies to the vortex elements generated at the rear of the disk, which remain always behind the disk and are only affected by the induced flow in this regime (Eq. 9) and the inflow μ_z. Equation 15 is applied here, with the only difference that the initial vertical position $z_v(x_e)$ is replaced by the vertical position of vortex release z_{vr}.

The wake geometry resulting from the models of Glauert, Lock, Drees and Beddoes in Fig. 5, based on the induced velocity distributions of Fig. 4, are computed for the case of no root cut-out $r_a = 0$. The most simple constant inflow assumption of Glauert is inappropriate since the wake is convected downward everywhere. This is in

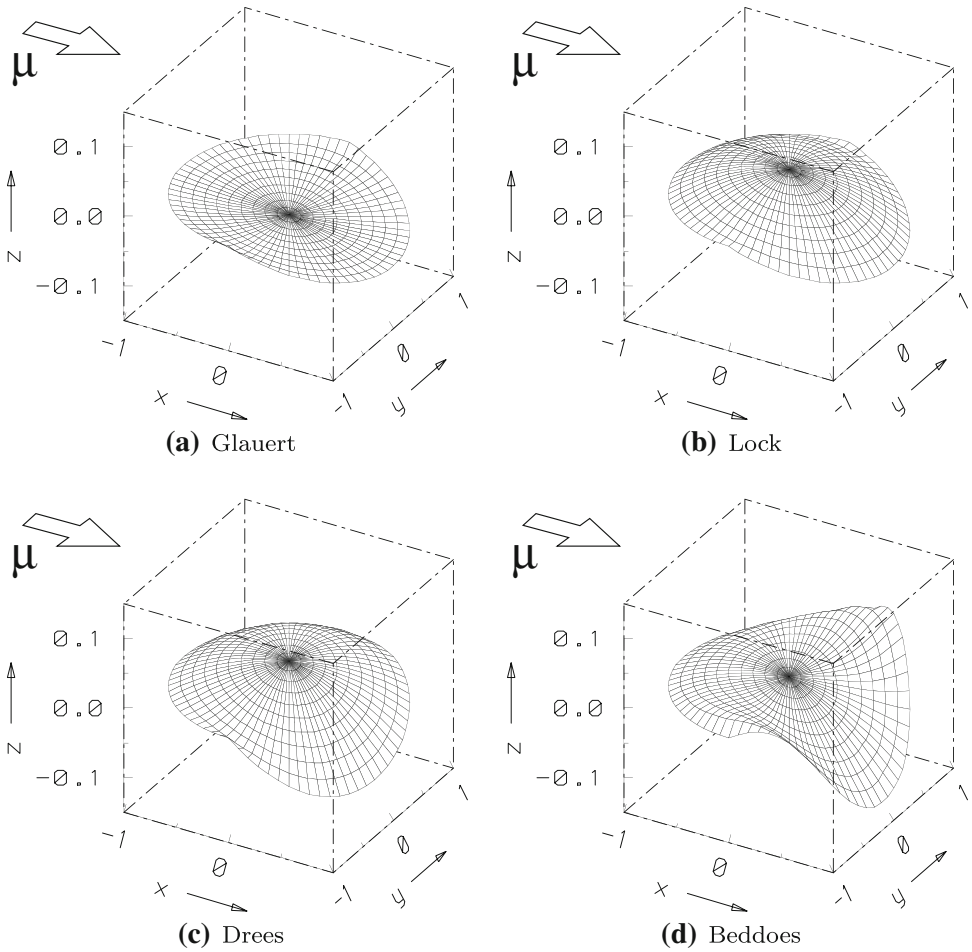


Fig. 5 Wake geometry of the various models, $r_a = 0$

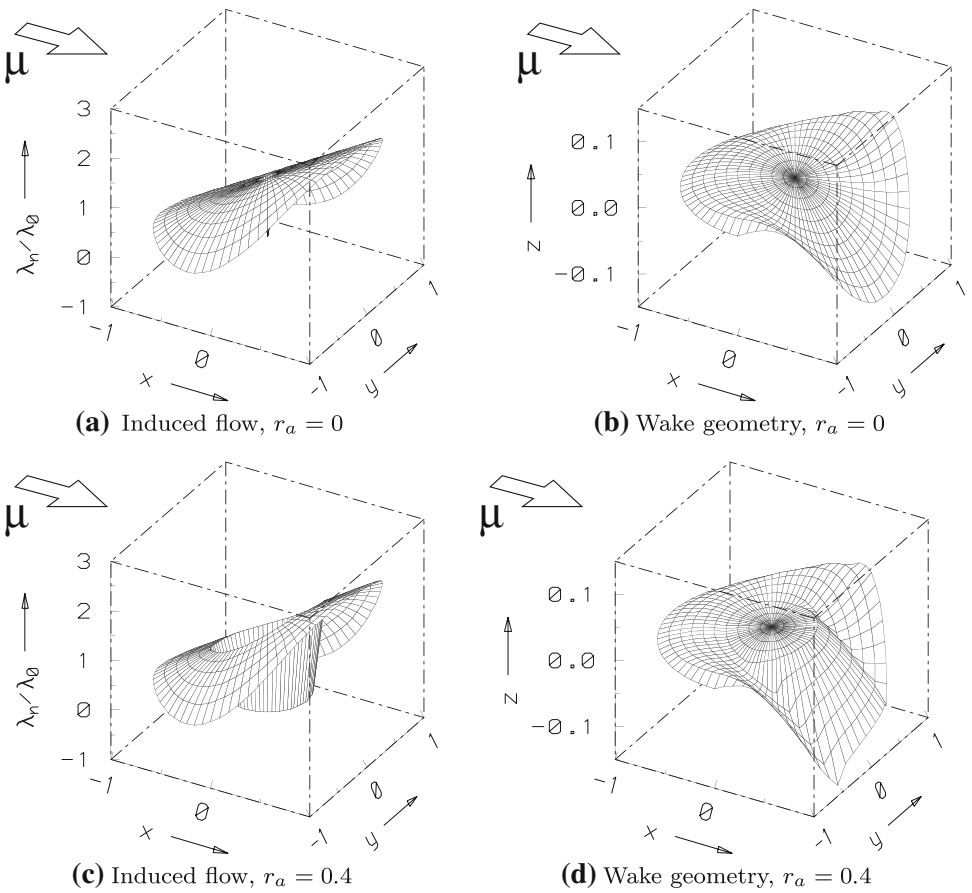
contrast to experimental observations that have shown an initial vertical upwards convection, followed by an increasing downward convection. This is due to upwash in the front of the disk and an increasing downwash towards the rear. It is qualitatively well represented by the Lock model, which has a longitudinal gradient of the induced flow. Drees model, in addition to the Lock gradient, has less downwash on the advancing and more on the retreating side, thus lifting the wake geometry on the advancing side and pushing it more downwards on the retreating side. Beddoes attempts to simulate the lateral roll-up by modifying the induced velocities exponentially increasing towards the sides proportional to $|y|^3$ (see Eq. 7), thus lifting the wake geometry there.

The effect of root cut-out is illustrated in Fig. 6 for the Beddoes model only; the modifications are of the same type in all the models. As can be seen, the root cut-out area has no induced flow, the momentum is applied to the lifting part of the disk only and thus any root cut-out increases the induced flow since the lifting area is reduced. The consequences on the wake geometry are seen in the cut-out area and downstream of it. In this operating condition with a positive tip path plane angle of attack, a part of the oncoming flow is penetrating the disk from below, thus pushing up the wake when no induced flow is opposing it, as is the case in the root cut-out area.

3 Extension to harmonic blade loading

As the steady thrust defines the mean induced flow by means of momentum theory it must be assumed as well that the azimuthally (and radially) varying distribution of the blade lift will cause an associated amount of varying induced flow. This is in analogy of the combined blade element/momentum theory, which is elaborated in most textbooks to the subject, like in [15, 16]. In the examples given therein it is applied to the hover case with constant azimuthal (i.e., rotationally symmetric) yet radially varying distribution of the blade's lift distribution. The momentum theory then is applied to each of the blade elements individually and independently, assuming that there is no interference between the elements. In analogy and with the same justification to this radial variation of induced flow there can be assumed the same for the azimuthally varying part of the blade lift.

In the case of constant operating conditions (for example: in forward flight) a rotor blade will—after all transients



(a) Induced flow, $r_a = 0$ **(b)** Wake geometry, $r_a = 0$

(c) Induced flow, $r_a = 0.4$ **(d)** Wake geometry, $r_a = 0.4$

Fig. 6 The effect of a root cut-out on the induced flow and the wake geometry, Beddoes model

have died out—experience air loads that will contain a full range of azimuthal harmonics acting on it. Every blade will have the same loading when it is located at the same azimuth position. Any local increase above the mean loading (which generates the thrust) will—by virtue of local momentum theory—generate a local increase of induced flow and any loading less than the mean will generate a reduction of induced flow there. The superposition assumption then leads to the identification of the harmonic part of the induced flow.

3.1 Rotor-induced velocities

In general, since the mean rotor thrust—in combination with the operating condition (advance ratio, angle of attack)—provides the mean induced velocity plus its gradients longitudinally (=1/rev cosine variation) and laterally (=1/rev sine variation) as described before, the harmonic blade loading must be analyzed for the frequency range at integer harmonics above this range, i.e., from 2/rev to 6/rev, which covers the range of frequencies of interest. The limitation to the range of 2−6/rev is justified, because the higher the frequency, the smaller the wave length. Furthermore, the lift amplitudes of the higher harmonics usually are significantly smaller than those of the lower harmonics. In addition, a vortex segment passing through the higher harmonics of the induced flow with the short wave lengths are deflected less due to the short time that they experience the induced up- or downwash. In combination with the small amplitudes of the high frequency downwash the frequency range of 7/rev and higher is assumed to be negligible.

The approach of computing the higher harmonic downwash distribution is as follows. First, the loading time history of the blade during its revolution must be analyzed for its harmonic content by means of a Fourier analysis for the range of interest, i.e., from 2/rev to 6/rev. This provides the amplitudes and phases for every harmonic

$$L_n(\psi) = L_n \cos\,(n\psi - \psi_n) \quad n = 2, 3, \ldots, 6 \tag{17}$$

where the phase of the nth harmonic is defined by the cosine and sine coefficients, $\psi_n = \arctan(L_{n\mathrm{S}}/L_{n\mathrm{C}})$. Since the induced flow is caused by the lift, the phase of the harmonics of the induced flow are the same as the phase of the generating harmonic of the lift. The magnitude of the harmonic downwash is defined by the magnitude of the lift by means of momentum theory:

$$\lambda_{\mathrm{inh}} = \sqrt{\frac{N_{\mathrm{b}} L_n}{2\rho\pi R^2 (\Omega R)^2}} \quad n = 2, 3, \ldots, 6 \tag{18}$$

Setting $n = 0$, the result for the mean induced inflow is obtained. The index h is a reminder that this is a hover condition ($\mu = 0$). In forward flight the magnitude of the harmonic downwash will decrease with the same decay function as given in Eq. 5 for the mean inflow

$$\lambda_{\text{in}}(\mu, L_n) = \frac{\lambda_{\text{inh}}^2}{\sqrt{\mu^2 + (\mu_z + \lambda_{\text{i0}})^2}} \quad n = 2, 3, \ldots, 6 \tag{19}$$

Next, an assumption for the radial distribution of this higher harmonic induced flow has to be made. It can be assumed with some justification that the higher harmonic lift distribution is increasing quadratically from the hub towards the tip of the blade due to the quadratic increase of dynamic pressure. Since the induced flow is proportional to the square root of the lift (see Eq. 18) this results in a linear radial distribution of induced flow. However, to simplify the model a radial distribution function of the generic form

$$f(r) = f_m r^m \quad m = 1, 2 \tag{20}$$

may be assumed. $m = 0$ would thus cause $f = 1$, a radially constant distribution, but this would cause a non-physical singularity in the rotor centre. $m = 1$ is a linear increase from root to the tip, and $m = 2$ represents a quadratic distribution. f_m is a factor computed such that the integral over the rotor disk is 1 for whatever order of polynomial m is used, in order to satisfy the momentum theory, or

$$\frac{1}{2\pi}\int_0^{2\pi}\int_0^1 (1) r \, \mathrm{d}\psi \, \mathrm{d}r \stackrel{!}{=} \frac{1}{2\pi}\int_0^{2\pi}\int_0^1 (f_m r^m) r \, \mathrm{d}\psi \, \mathrm{d}r \tag{21}$$

This function $f(r)$ thus may be viewed as a non-dimensional radial mode shape with the amplitude of $\lambda_{\text{in}} = \sqrt{\lambda_{\text{inC}}^2 + \lambda_{\text{inS}}^2}$. Finally, the description of the higher harmonic downwash field then is

$$\lambda_i(r, \psi) = f(r) \sum_{n=2}^{6} \lambda_{\text{inS}} \sin n\psi + \lambda_{\text{inC}} \cos n\psi \tag{22}$$

The sum of Eqs. 7 and 22 then represents the total induced velocity field inside the rotor disk which is used for the wake geometry computation:

$$\lambda_i(x, y) = \lambda_{i0}\left[k_0 + k_x\left(x - |y|^3\right) + k_y y\right] + f(r) \sum_{n=2}^{6} \lambda_{\text{inS}} \sin n\psi + \lambda_{\text{inC}} \cos n\psi \tag{23}$$

An example of the relative geometric distribution of induced flow within the rotor disk for the harmonics $n = 2$,3 and 6 is given in Fig. 7 for both the cosine and the sine components. In these figures, the symbol λ_n/λ_0 is employed, where $\lambda_n = \lambda_i(r, \psi)$ and $\lambda_0 = \lambda_{\text{in}}$. The absolute values are then obtained by multiplication with the magnitude, depending on the harmonic content of blade loading as identified from the simulation.

3.2 Computation of the wake geometry

Like the wake geometry computation in the previous section, the wake convection is performed in downstream direction, i.e., along x. Therefore, the higher harmonic downwash distribution must be transformed from the polar coordinates (radial polynomial in $f(r)$ and Fourier series in azimuth) into Cartesian coordinates, which for the Fourier series requires the application of Moivre's formulae. Any higher harmonic $\cos n\psi$ and $\sin n\psi$ can be expressed in terms of powers of $\cos \psi$ and $\sin \psi$, and these in terms of x/r and y/r, for example see Eq. 11.

In the following, the second harmonic ($n = 2$) is elaborated, which results in simpler formulae. All higher harmonics follow the same procedure with increasing complexity in the resulting expressions. Again, the radial distribution is assumed to be linear ($m = 1$) for the same reason of simplicity, $f(r) = f_1 r$. Thus,

$$\cos 2\psi = \cos^2\psi - \sin^2\psi \qquad \sin 2\psi = 2 \sin\psi \cos\psi \tag{24}$$

Making use of $\cos\psi = x/r = x/\sqrt{x^2 + y^2}$ and $\sin\psi = y/r = y/\sqrt{x^2 + y^2}$ it follows that

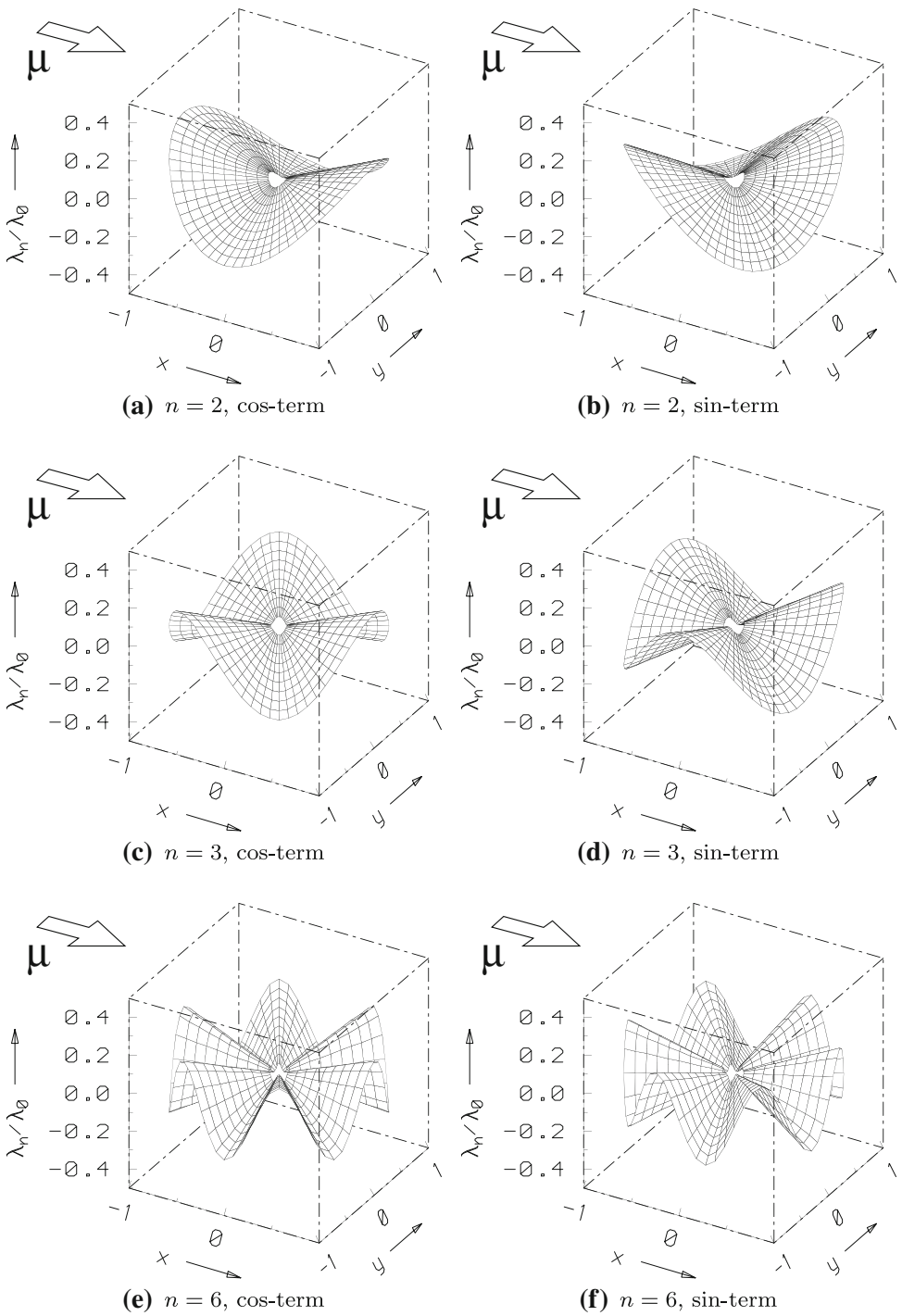
$$\begin{aligned}\lambda_{\text{i2}}(x, y) &= f(r)(\lambda_{\text{i2C}} \cos 2\psi + \lambda_{\text{i2S}} \sin 2\psi) \\ &= f_1 \lambda_{\text{i2C}} \frac{x^2 - y^2}{\sqrt{x^2 + y^2}} + 2 f_1 \lambda_{\text{i2S}} \frac{xy}{\sqrt{x^2 + y^2}}\end{aligned} \tag{25}$$

Now that the higher harmonic induced velocity field is expressed in cartesian coordinates, the remainder of the analysis is identical to what was shown in the previous section. The additional vertical wake distortion is obtained by analytical integration from the point of vortex release to the point of interest within the disk:

$$\begin{aligned}\Delta z_v(x, y) = -\frac{1}{\mu}\int_{x_{\text{vr}}}^{x} \lambda_{i2}(x, y) \, \mathrm{d}x &= -\frac{f_1 \lambda_{\text{i2C}}}{\mu}\int_{x_{\text{vr}}}^{x} \frac{x^2 - y^2}{\sqrt{x^2 + y^2}} \, \mathrm{d}x \\ &- \frac{2 f_1 \lambda_{\text{i2S}}}{\mu}\int_{x_{\text{vr}}}^{x} \frac{xy}{\sqrt{x^2 + y^2}} \, \mathrm{d}x\end{aligned} \tag{26}$$

This applies to all vortex segments generated in the front of the disk, i.e., for $x_{\text{vr}} < 0$. All others are generated at the rear of the disk and are traveling away from it, and hence are not affected by the harmonic loading within the disk. The kernel of the integrals can readily be obtained using standard mathematical textbooks and integral tables (for example, [19]) or a symbolic computer software. The three types of integrals in Eq. 26 are

$$\int \frac{x^k}{\sqrt{x^2+y^2}}\,\mathrm{d}x = \begin{cases} \ln\,(x+\sqrt{x^2+y^2}) & k=0 \\ \sqrt{x^2+y^2} & k=1 \\ (x/2)\sqrt{x^2+y^2} - (y/2)\,\ln\,(x+\sqrt{x^2+y^2}) & k=2 \end{cases} \tag{27}$$



(a) $n = 2$, cos-term **(b)** $n = 2$, sin-term

(c) $n = 3$, cos-term **(d)** $n = 3$, sin-term

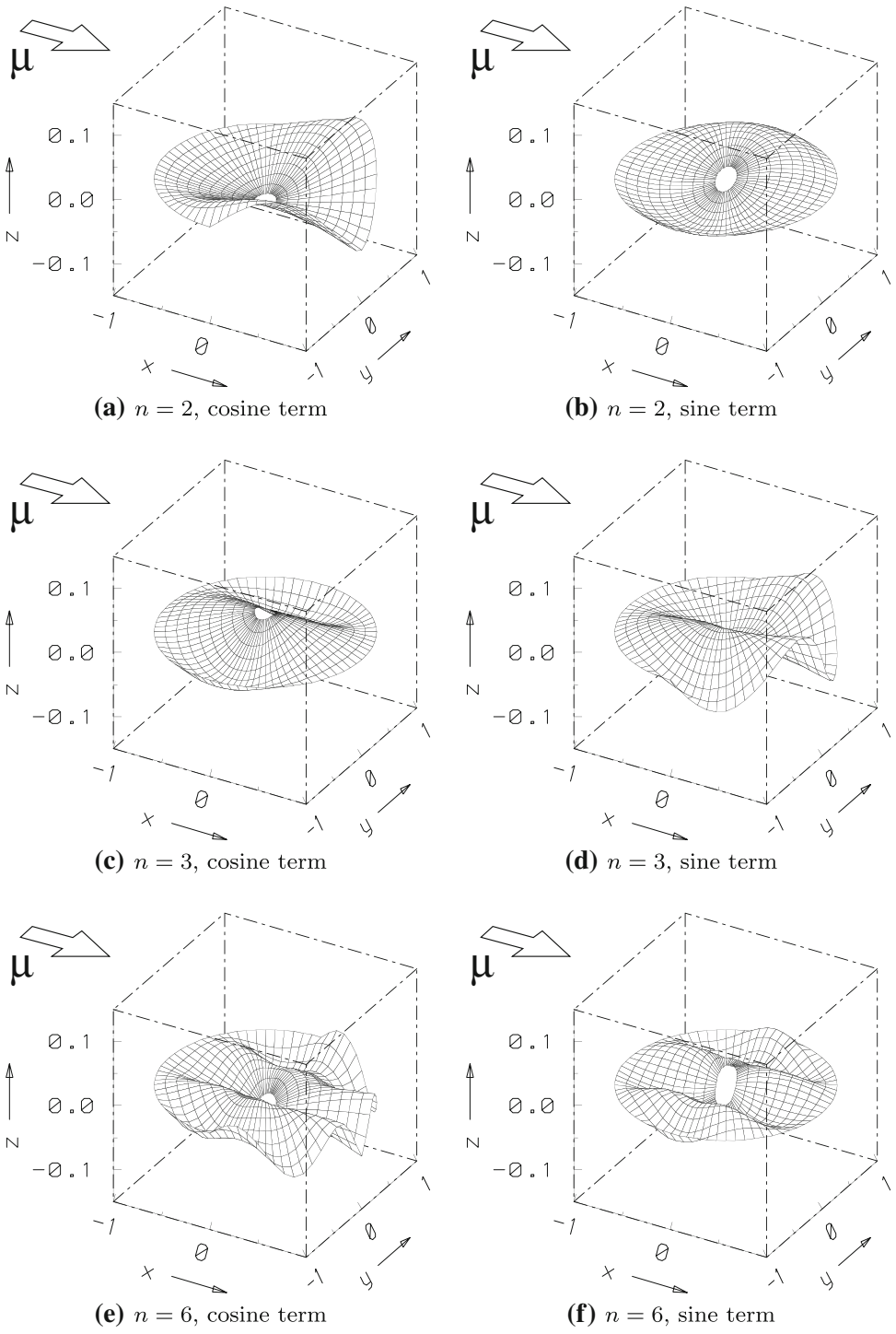
(e) $n = 6$, cos-term **(f)** $n = 6$, sin-term

Fig. 7 Distribution of induced velocity for the harmonics 2, 3 and 6/rev; radially linear ($m = 1$)

With the use of these formulae, at every point within the rotor disk the vertical deflection of the wake can be computed. The results are shown in Fig. 8 for the harmonics 2, 3 and 6, based on the induced velocity distributions of Fig. 7. As can be seen, the higher the harmonic, the less the wake deflects. This is due to the smaller spatial wavelength at the higher harmonics, when compared with the lower harmonics.

3.3 The HART II minimum noise and minimum vibration cases

Within HART II [24], two major settings of higher harmonic control (HHC) were investigated, both with application of a blade root pitch amplitude of 0.8° at 3/rev. The minimum noise (MN) case had a phase of $\psi_3 = 300°$, while the minimum vibration case (MV) had a phase of



(a) $n = 2$, cosine term **(b)** $n = 2$, sine term

(c) $n = 3$, cosine term **(d)** $n = 3$, sine term

(e) $n = 6$, cosine term **(f)** $n = 6$, sine term

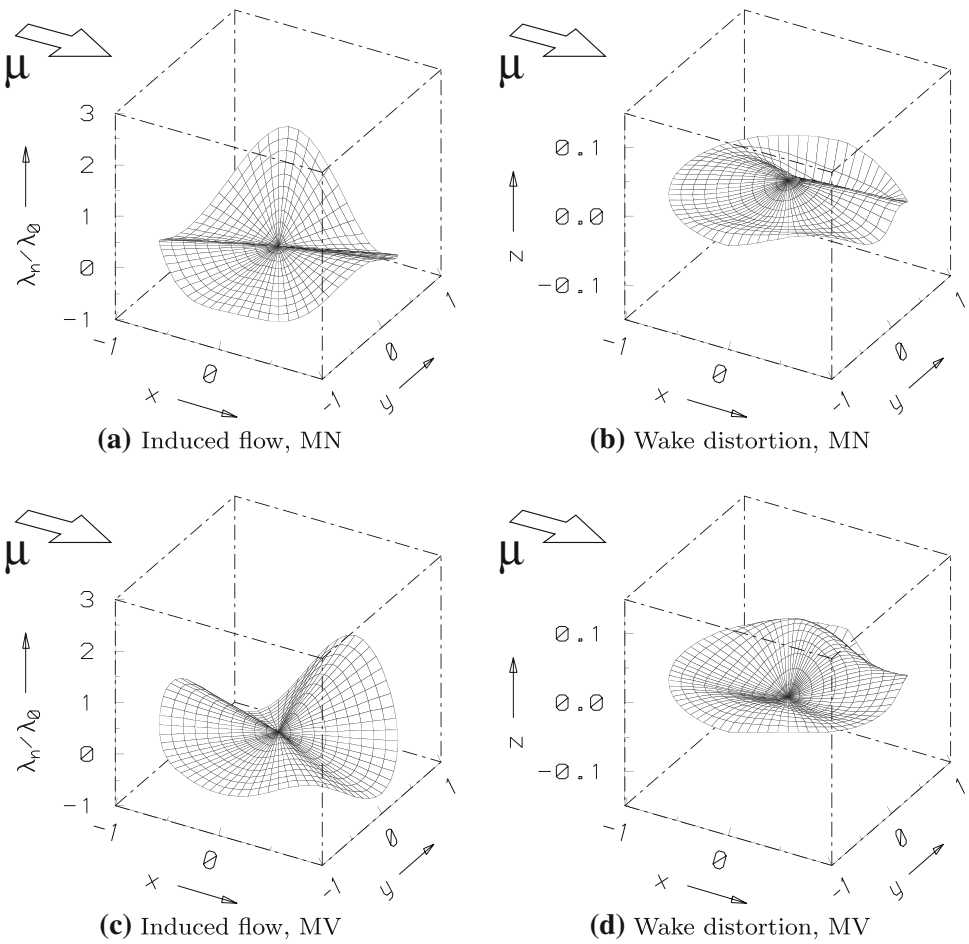
Fig. 8 Wake distortion due to the harmonics 2, 3 and 6/rev; based on the induced flow of Fig. 7

$\psi_3 = 180°$. These 3/rev HHC cause a strong 3/rev content of the blade loading, and due to interharmonic coupling in forward flight, a significant increase of the 2/rev blade loading also. The other harmonics (4–6/rev) were affected as well, but less significantly. For this paper, the magnitudes and phases of the blade loading in the range from 2/rev to 6/rev were extracted from validated simulations of these cases, and the associated induced inflow amplitudes computed using the method described in the section above. The superposition of all these harmonics and their phases then leads to the induced flow distributions shown in Fig. 9, which cause the wake deflections given therein as well.

It must be noted that this is only the part due to the harmonics 2–6/rev and thus adds on top of the underlying basic induced flow (Figs. 4 or 6), and the deflections of Fig. 9 add on top of the underlying basic wake geometry as shown in Fig. 5 or Fig. 6. First, have a look at the MN case. The induced flow Fig. 9a has a large increase in the second quadrant of the disk. Any vortex element passing this area consequently gets a downward convection as can be seen well in the associated wake distortion Fig. 9b on the advancing side, especially in the region of $y \approx 0.7$, which is relevant for BVI in the first quadrant. The vortex position measurements performed during the HART II test confirm exactly this behaviour of the wake. Second, the different HHC pitch control phase of the MV case generates a different loading distribution of the blade (compared to the MN case) and thus the associated induced flow field looks different as well.

Due to all of the non-linearities of the forward flight aerodynamics of a helicopter rotor, the loading and induced flow distribution are not just rotated in azimuthal direction by the difference of the HHC control phases. The individual magnitudes are also observed to be modified to a certain degree. However, the dominant effect is seen primarily in the rotation (compare Fig. 9c with a). This induced flow field of the MV case Fig. 9c causes the wake distortions as shown in Fig. 9d. Note that for simplicity in all these figures the tip vortex is assumed to be emitted from the blade tip.

In any HHC case, however, there is a region of download at the tip, resulting in double vortex systems with a counter-rotating vortex at the blade tip, and another (conventionally rotating) vortex at further inboard location of $r \approx 0.85$. These double vortex systems were confined to a limited azimuthal extent only. Even in cases without active



(a) Induced flow, MN **(b)** Wake distortion, MN

(c) Induced flow, MV **(d)** Wake distortion, MV

Fig. 9 Induced flow and associated wake distortion for the HART II minimum noise (MN) and minimum vibration (MV) cases, harmonics 2–6/rev, radially linear inflow ($m = 1$)

control such events were frequently noticed, especially in high speed forward flight, where in the second quadrant the tip may encounter negative loading associated with the generation of such double vortex systems. The vortex trajectory with a vortex release point $r_{vr} < 1$ will therefore not capture the full distortions as shown in the figures since it does not include the induced flow field upstream of its release point. However, this does not cause any problem for the mathematics of this approach. The lower boundary of the integrals for the wake deflections must be set to the actual release coordinate x_{vr} as identified from the radial loading distribution.

4 Accounting for the fuselage

The flow around the fuselage causes disturbances within the rotor disk since the fuselage and rotor usually are close to each other. Fuselage proximity is commonly acknowledged to be a prime cause of rotor vibration, but a large rotor-fuselage separation distance requires a long rotor shaft (thus increasing the empty mass of the vehicle) and simultaneously may cause problems with respect to ground resonance due to a reduced stiffness of the hub attachment to the fuselage. However, the effect of flow induced by the fuselage on rotor blade aerodynamics often is ignored in comprehensive simulation codes and even more rarely included in prescribed and free-wake models.

To account for the fuselage displacement flow, either a panel method or CFD code must be applied in order to obtain the distribution of fuselage-induced velocities on the rotor disk. In general, the flow field is fully three-dimensional with large gradients in all directions close to the fuselage, and a non-linear velocity distribution will be obtained in the rotor disk. In a cylinder of a few chords in length above and below the rotor disk such gradients will still prevail, but it can be assumed that the gradients in longitudinal (x) and lateral (y) are much larger than in the vertical (z) direction, since within this cylinder the extension in x and y is much larger than in z. Therefore, to simplify the problem and to reduce it to the fundamental underlying physics, an assumption can be made that within this cylinder (wherein all the vortex filaments are traveling downstream) no gradients in z direction are present and only the variation of fuselage-induced velocities normal to the disk are taken into account. However, the validity of this assumption must be verified by computing the velocity field in the volume of interest and analyzing it for the gradients.

4.1 Fuselage-induced velocities

In the incompressible case, the fuselage-induced velocities are linearly proportional to the flight speed, which is in contrast to the rotor-induced velocities caused by rotor thrust, which based on momentum theory approach zero for very high speed. Since helicopters do not generally fly faster than a Mach number of 0.3, the assumption of incompressible flow over the fuselage is valid. Therefore, the application of an incompressible panel method is justified to approximate the fuselage-induced velocities that will impinge and influence the rotor disk. This has been performed in [20]. The result was formulated in terms of polynomials of third order in radial direction and a Fourier series up to the seventh harmonic in azimuthal direction for convenient usage in a comprehensive rotor simulation environment. An example of the flow induced by the wind tunnel model fuselage is given in Fig. 10. Only cos $n\psi$ terms appear due to the lateral symmetry of the fuselage. Yet, this model has not been used so far for affecting the prescribed wake's geometry, which is addressed in this section.

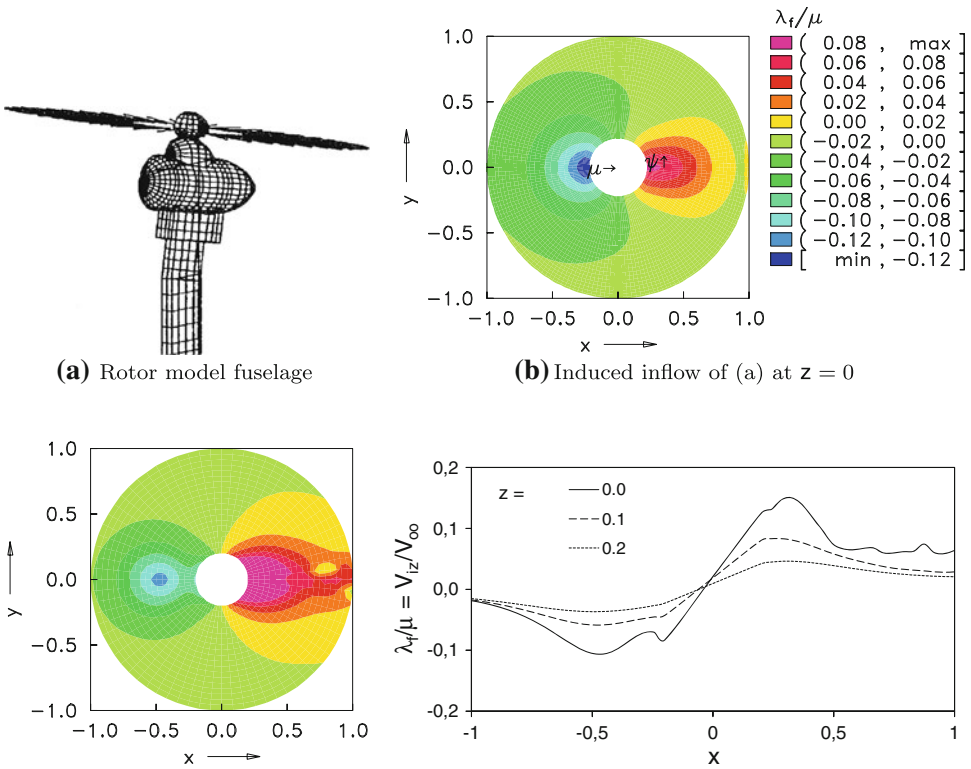
The normal velocities in the rotor plane ($z = 0$) of two different fuselages are shown: in Fig. 10a a fuselage as used on a vertical strut in the wind tunnel is modeled by a panel method, and in (b) its induced velocity field in the plane of the rotor disk at $z = 0$ is shown. The fuselage used in the HART II test mounted on the rear sting support is shown in Fig. 1 and is computed by a Navier–Stokes code. In (c), the fuselage-induced velocities of this HART II fuselage are shown for comparison to (b) and in (d) the velocities along the centreline at $y = 0$ are given for different heights above the rotor disk with increments of 10% of the radius (=1.65 chord lengths). Again the data are extracted from a Navier–Stokes computation. As can be seen, there is a strong effect of distance to the fuselage, and a generic model should include these. For small variances around $z = 0$, however, this effect may be ignored.

Since the fuselage-induced flow is described similar to the induced velocities due to harmonic blade loading (combination of Fourier series and radial polynomial functions) the same method can be applied here as well. However, although the description of [20] could be used directly it is more desirable here to take advantage of a more simple and generic description, to elaborate the procedure.

The radial polynomial functions may be seen as mode shapes $\phi_n(r)$ of the fuselage-induced inflow and the Fourier series $q_n(\psi)$ represents the magnitudes and circumferential shape. Then,

$$\frac{v_{if}}{V_\infty} = \sum_{n=1}^{N} \phi_n(r) q_n(\psi) \tag{28}$$

where the mode shape functions also depend on the fuselage angle of attack:



(a) Rotor model fuselage

(b) Induced inflow of (a) at z = 0

(c) Induced inflow of HART II fuselage (see Fig. 1) at z = 0

(d) Distribution at the centreline, HART II fuselage

Fig. 10 Fuselage-induced velocities in the rotor disk, wind tunnel model. The CFD data shown in **b** and **c** are provided by Konkuk University, Seoul, Korea (unpublished)

$$\phi_n(r) = \sum_{j=0}^{J} c_{nj}(\alpha) r^j \tag{29}$$

and j, $J \geq 0$ or j, $J \leq 0$, depending on the distribution of velocities obtained by panel or CFD method, or by measurements. The Fourier series becomes

$$q_n(\psi) = \sum_{k=0}^{K} a_{nk} \cos k\psi \tag{30}$$

For simplicity, in order to show the method, a simplified version is used and may be classified as a generic case. It is assumed that the fuselage effect can be represented by the following polynomial function:

$$\phi(r) = \begin{cases} c_0 + c_1 r_a/r & r \geq r_a \\ c_0 + c_1 r/r_a & r < r_a \end{cases} \tag{31}$$

This distribution is a representative approximation to the distribution shown in Fig. 10c for $z = 0$ and features a reduction of fuselage influence outside a radius r_a (which may conveniently be chosen as the root cut-out of the blades) inversely proportional to the distance (and asymptotically approaching zero for very large distances as required), in analogy to the induced velocities of a vortex outside its core radius. Inside this radius r_a, a linear variation proportional to the distance from the hub centre (and usually the fuselage centre line as well) is assumed to have zero influence in the centre itself. Here, the longitudinal point of symmetry on the fuselage is assumed, and thus no vertical flow can be induced by the fuselage. For the Fourier series, again due to simplicity, only the first harmonic is considered:

$$q(\psi) = \cos \psi = x/r \tag{32}$$

Note that $r = \sqrt{x^2 + y^2}$. Next, assume $c_0 = 0$ due to symmetry of the fuselage in the front half of the disk and in the rear half of it, such that the sum of all upwash in the front must be compensated by the sum of all downwash in the rear. This assumption is only made for simplicity of this demonstration model, it is by no means general nor mathematically necessary and a realistic fuselage will not reflect this. However, the fundamental feature of an upward-downward flow from front to rear will be present in all fuselage flow fields and this is represented by the model. Also, $c_1 = 1$ is used as a generic magnitude, which in a real application must be identified from the panel code or CFD computation. Therefore,

$$\frac{v_{if}(r,\psi)}{V_\infty} \approx \frac{\lambda_f}{\mu} = \phi(r) q(\psi) = \begin{cases} r_a x/(x^2 + y^2) & r \geq r_a \\ x/r_a & r < r_a \end{cases} \tag{33}$$

4.2 Computation of the wake geometry

As in the case of the previous sections for the computation of the basic wake geometry as well as for the distortions due to harmonic loading, only the vortex elements generated in the front of the disk will pass the fuselage-induced velocity field. The distortion caused by the induced flow given in Eq. 33 is again computed by integrating along the way traveled through this velocity field:

$$z = -\frac{1}{\mu}\int_{x_{\mathrm{vr}}}^{x} \lambda_f(x,y)\mathrm{d}x = \begin{cases} -(r_\mathrm{a}/2)\,\ln\,(x^2+y^2) & r \geq r_\mathrm{a} \\ -x^2/(2r_\mathrm{a}) & r < r_\mathrm{a} \end{cases} \tag{34}$$

As can readily be seen, the wake distortion is not a function of the advance ratio, i.e., it is independent of the flight speed. This is appropriate to the physics since the induced velocities are increasing with flight speed, but simultaneously the downstream convection speed of a vortex element is increasing by the same amount, or, the time it is subjected to this velocity field is inversely reducing. In summary, both effects cancel each other and the distortion of the wake geometry due to presence of the fuselage is independent of flight speed.

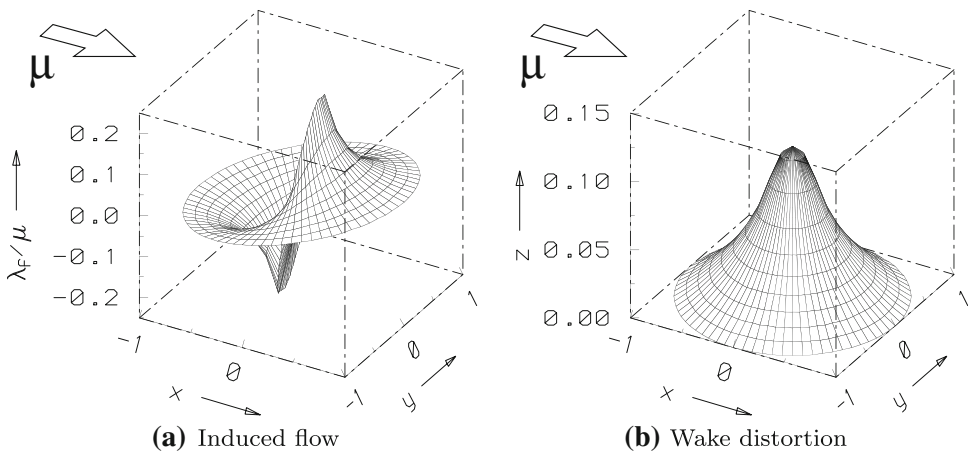
An example for the effect of this generic fuselage displacement flow model is illustrated in Fig. 11. In (a), the fuselage-induced flow field is shown, with zero at $x = 0$ in the symmetry line. From the root cut-out towards the edge of the disk the proportionality to $1/r$ is clearly visible, with maximum values at the fuselage symmetry line for $y = 0$ and quickly reducing in magnitude in lateral direction, representing the small width of a typical helicopter fuselage. In the longitudinal direction at $y = 0$, the flow disturbance is larger. The associated wake distortion is given in Fig. 11b. The largest vertical distortion is found at the hub centre. At the rear rim of the disk, all fuselage-induced distortions vanish, which represents the sum of all induced velocities encountered from the front to the rear is zero, as said before.

To account for varying distances of the rotor disk relative to the fuselage or of vortex elements relative to the plane $z = 0$, the induced velocities may be formulated as a function of z, which requires the extraction of the fuselage induced flow field from the CFD data at different values of z. Although this complicates the numerical procedure of computing vortex trajectories it is no problem in general and this is left for future investigations. In general, the fuselage-induced flow vanishes for large distances to it and therefore the vortex trajectory distortions due to the fuselage vanish as well.

The symmetry assumption to the fuselage induced flow field does not hold also in tandem rotor helicopters. The front rotor experiences a strong upwash in the front but only little downwash in the rear while the rear rotor will get a weak upwash in its front and a large downwash in the rear. These modifications can all be dealt with in the analytical modeling and only require an appropriate set of analytical functions representing such velocity fields. Again, this is left for future investigations.

5 Application and validation

A first attempt of applying higher harmonic elements to the induced flow and computing the associated wake distortions was made in 2000 [18]. At that time a different approach was used: in analogy to the basic (static) wake, which is based on the steady operational conditions, the dynamic characteristics of the rotor were used to calculate the dynamic part of the induced flow when HHC is applied. This was achieved by identification of a transfer function between the HHC input at the blade root and the dynamic response of the rotor blade in flap and torsion, thus providing the magnification and phase lag of HHC-related



(a) Induced flow **(b)** Wake distortion

Fig. 11 Fuselage-induced velocities in the plane $z = 0$ and their influence on wake distortion

airloads relative to the HHC input. In general this approach is valid and allows the wake distortion to be computed prior to the rotor simulation, but limited to HHC applications. In the presence of harmonic air loads not due to HHC (e.g., in fast forward flight), this method fails to predict any wake distortion. Another drawback of this approach is that the dynamic behaviour of every rotor to be computed needs to be analyzed for the transfer function between HHC input and blade dynamic response in flap and torsion, which in addition will be dependent on the flight condition. Thus, that model was too limited for a wider application.

The method presented here was developed to iterative adapt itself to the operating condition. No knowledge is required about the dynamic transfer function of the rotor blade, and it does not matter which of the following list of physical effects caused the dynamic blade loading: HHC, IBC, active flaps, active twist, the operating condition by itself or an arbitrary combination of all together. First applications and validations were presented in 2007 [21] and 2009 [22, 23], using the HART II [24] and HART I [25] data.

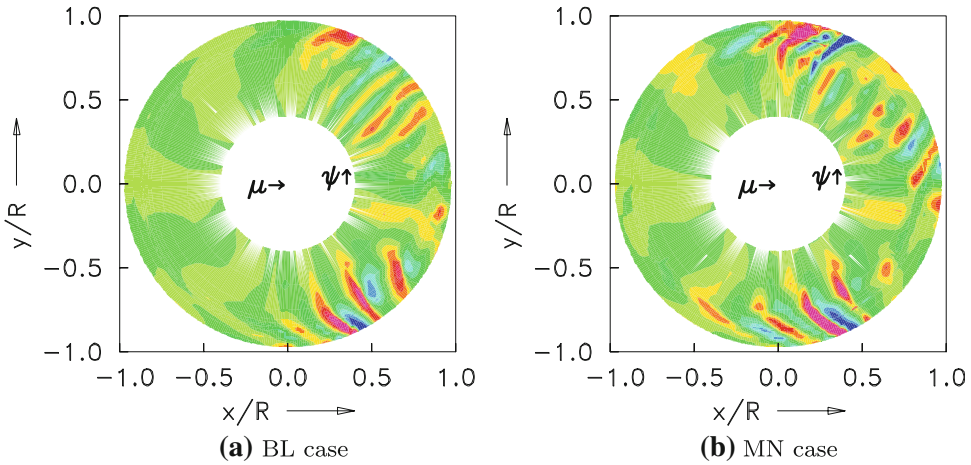
The comprehensive rotor simulation code of DLR (S4), based on blade element theory, is applied to test the modifications. It uses advanced models for the section aerodynamics (semi-empirical math model for two-dimensional air loads, including: unsteady, compressibility, dynamic stall and yaw flow effects); the computation of induced velocities (tip loss, prescribed wake, multiple trailers, accounting for root cut-out, but no fuselage influence included so far); and the elastic blade dynamics are represented by means of a series of modes in flap, lead-lag and torsion and their associated natural frequencies. A trim routine modifies the control inputs until the desired operational condition is obtained, and all transients have died out. Acoustic post-processing was performed by the APSIM code [26] with the section loading and the blade motion of the S4 results.

A wide range of validation cases has been performed with respect to blade motion (in terms of the static and dynamic tip deflection in flap and torsion), section loading at 87% radius with respect to the lower harmonics and especially the high frequency content caused by blade–vortex interaction (BVI). The qualitative location of most intense BVI within the rotor disk, the vortex trajectory at two lateral positions of $y = \pm 0.7$, and finally the noise radiation are also available.

As an example, the application of the wake improvements is demonstrated to assess the difference between the HART II baseline case (BL) and the minimum noise case (MN) was presented in [22]. The operational condition is the same for both cases as noted before. In the BL case no HHC input was applied, while in the MN case a 3/rev HHC with a blade root control amplitude of 0.8° and a phase of 300° was introduced. A 3/rev air load variation occurred naturally on the blade, with some increase of the 2/rev and the 4/rev due to inter-harmonic coupling in forward flight. In Fig. 12 the leading edge pressure distribution for these cases is given and is high-pass filtered at 10/rev in order to make the blade-vortex interaction (BVI) locations visible.

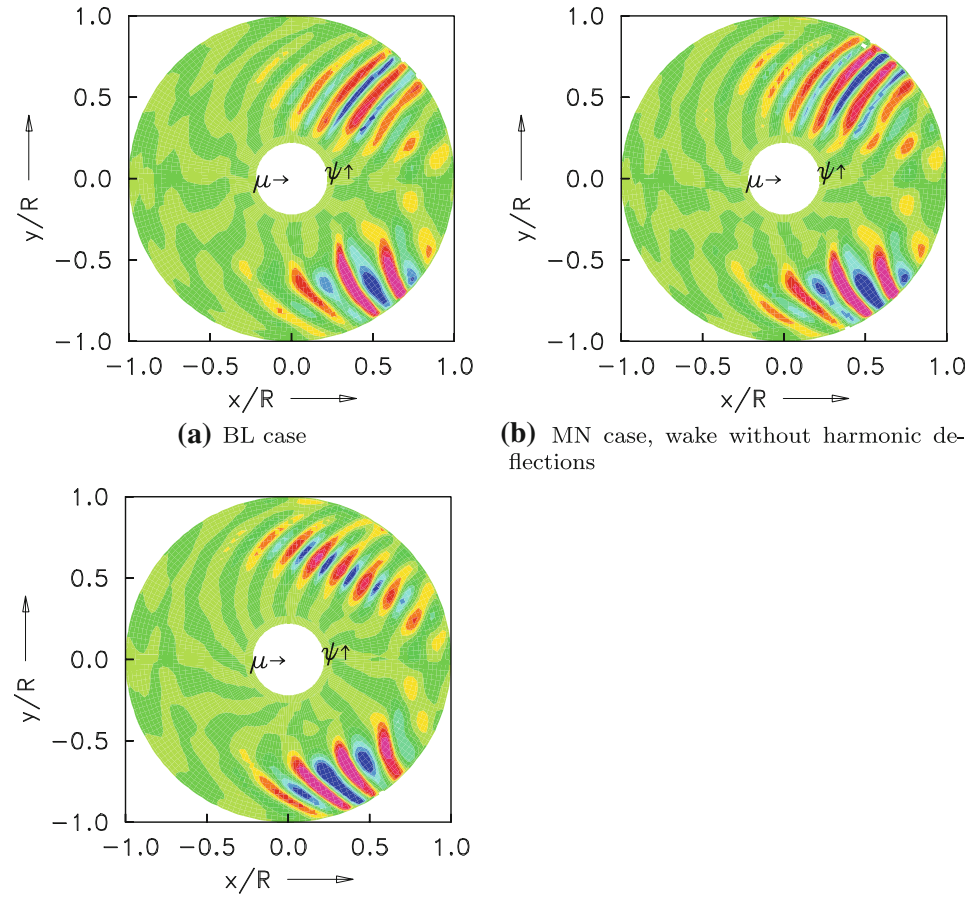
Comparing the MN case Fig. 12b to the BL case (a), the BVI locations in the first quadrant of the disk have moved inboard and towards the front of the disk, where the interaction angle is far from parallel, thus reducing noise. The location in the baseline case at about 50° azimuth, where parallel BVI occurred, is free of BVI in the MN case. This is caused by a larger blade lift (due to HHC and the associated blade response in torsion and flapping) in the second quadrant, generating more downwash in this region than in the BL case. Tip vortices traveling across this downwash are convected downward and thus pass through the rotor disk further upstream than in the BL case. On the retreating side, the BVI locations also have moved towards the front of the disk, again due to a larger blade loading around 230° azimuth, caused by HHC and the blade aeroelastic response. The simulation of these two cases is given in Fig. 13 in form of the blade loading C_nM^2, which is again high-pass filtered at 10/rev, allowing a qualitative comparison of BVI locations and relative strength with the experimental data of Fig. 12. First, the Beddoes' model (including the correction for the mean downwash) is applied to the BL case, Fig. 13a. A good overall agreement can be seen with the experiment for both the advancing and the retreating side BVI. Since the operating condition (trim) of the rotor was unchanged when HHC was applied in the MN case, the Beddoes' model does not change its geometry, see Fig. 13b, except for the blade motion, which is different from the BL case and cannot directly be seen in the figure. However, the flapping is very limited due to centrifugal forces.

The vortex circulation strength changes significantly according to the blade air loads. As a consequence, the BVI locations remain virtually unchanged, but the strength at the interaction location has increased due to the strong vortex circulation caused by HHC. Comparing Fig. 13a with (b), an increase of noise radiation is the immediate consequence, but in direct opposition to what the experiment has shown. This effect has caused the rotorcraft community to judge prescribed wake codes as inappropriate for active control conditions. Next, the application of the wake deflections due to harmonic blade loading results in the BVI locations shown in Fig. 13c. Compared to the BL case in (a), the BVI locations have shifted significantly inboard in the first quadrant and towards the front of the disk. This is in good qualitative and—with respect to the geometry—quantitative agreement with the experiment shown in Fig. 12b. On the retreating side, the BVI



(a) BL case **(b)** MN case

Fig. 12 Blade leading edge pressure distribution showing BVI locations. HART II experimental data, 10/rev high-pass filtered



(a) BL case

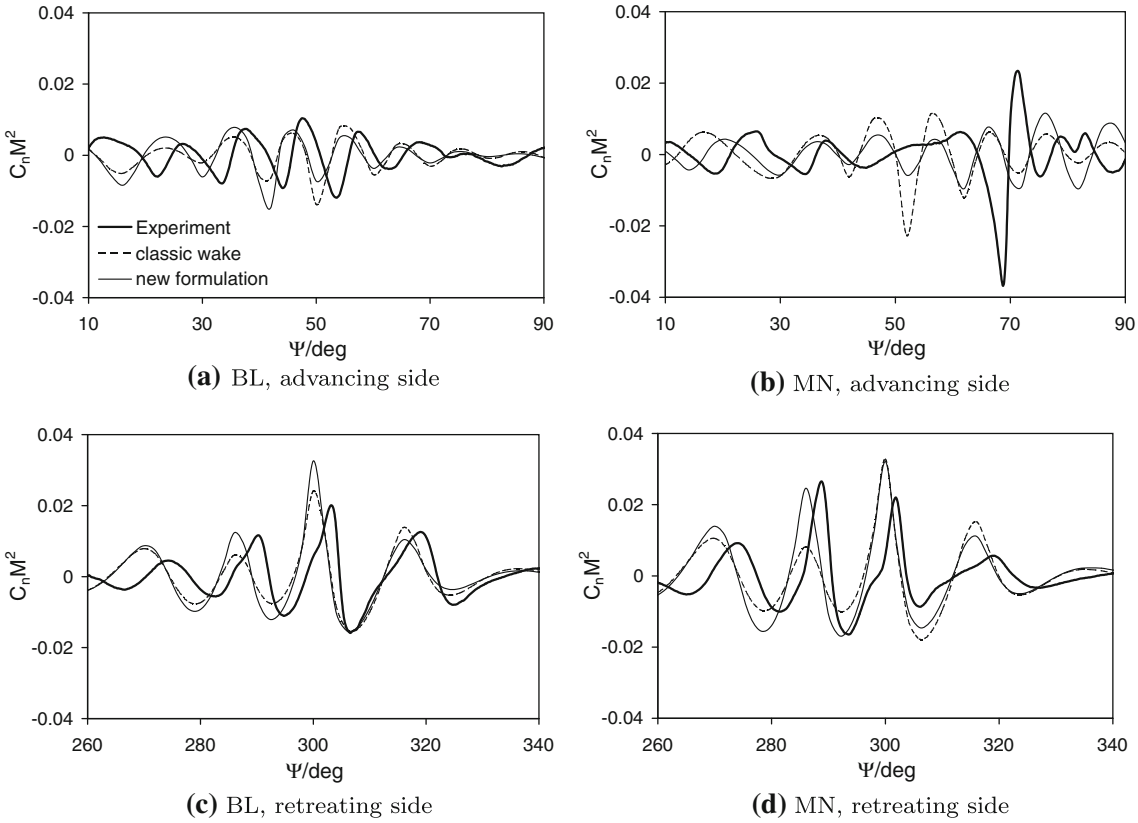
(b) MN case, wake without harmonic deflections

(c) MN case, wake including harmonic deflections

Fig. 13 Blade loading distribution showing BVI locations. Simulation, 10/rev high-pass filtered

locations have moved forward toward the front of the disk, also in agreement with the experiment in Fig. 12b. The reason for these improvements is due to the fact that the underlying physics of how harmonic airloads affect the vortex trajectories are now taken into account to a first order of accuracy. At least in this case it allows the application of prescribed wake codes for more complex conditions than before, and it is assumed that the formulation is general enough for the majority of operational cases.

This is further demonstrated in the time history of the high frequency content of the aerodynamic loading at a non-dimensional radius of $r = 0.87$, where pressure measurements along the chord are available from the



(a) BL, advancing side

(b) MN, advancing side

(c) BL, retreating side

(d) MN, retreating side

Fig. 14 Section loading at BVI locations, 10/rev high-pass filtered

experiment. These were integrated to compute the section loading C_nM^2, which can be compared to the simulation using the classic wake and the wake including harmonic deflections. In the BL case in Fig. 14a, c only small effects are seen from the inclusion of the harmonic wake deflection, which is due to the small amount of harmonic loading content in this case. The MN case, however, has a strong 3/rev loading that causes large wake perturbations. These affect the section loading high frequency BVI content, as clearly seen in (b) and (d), especially on the advancing side (b). The classic wake without these perturbations shows the location of the strongest BVI at about the same azimuth as in the BL case. Since the vortex strength due to the increased blade loading at the azimuth of vortex generation is also increased, the BVI intensity in the MN case therefore is stronger than in the BL case. Actually this is the case in the experiment as well, but at a significantly different azimuthal position around 70°, predicted correctly by the wake including the harmonic deflections. The effects on the retreating side are less pronounced, but again the wake including the harmonic deflections generates results closer to the experiment than the classic wake.

A further validation is given using the measured wake position data, compared to the predicted wake positions (see Fig. 15). It can be clearly seen that the classic wake tip vortex trajectory is almost the same in the BL and the MN cases, except for minor differences of the blade motion at the left side where the vortex is released. The trajectories of the new formulation including the harmonic wake deflections are significantly different, and with respect to the shape of the trajectory, they are much closer to the experimental data—a further indication of the correct interpretation of the underlying physics.

6 Discussion

In general, a large improvement is achieved when using the harmonic inflow model and the associated wake distortions. The cause of this improvement is identified from the wake distortions due to the dynamic part of the loading, which is most apparent when active control is applied. In these cases, a classic prescribed wake model does not change at all since the general operating condition remains unchanged. The only difference of active control cases to the baseline case without active control is due to a different blade motion which only slightly modifies the BVI zones.

Yet the majority of active control effect on noise radiation is not due to a modified blade motion but is due to a modified wake geometry, which is ten times larger than the

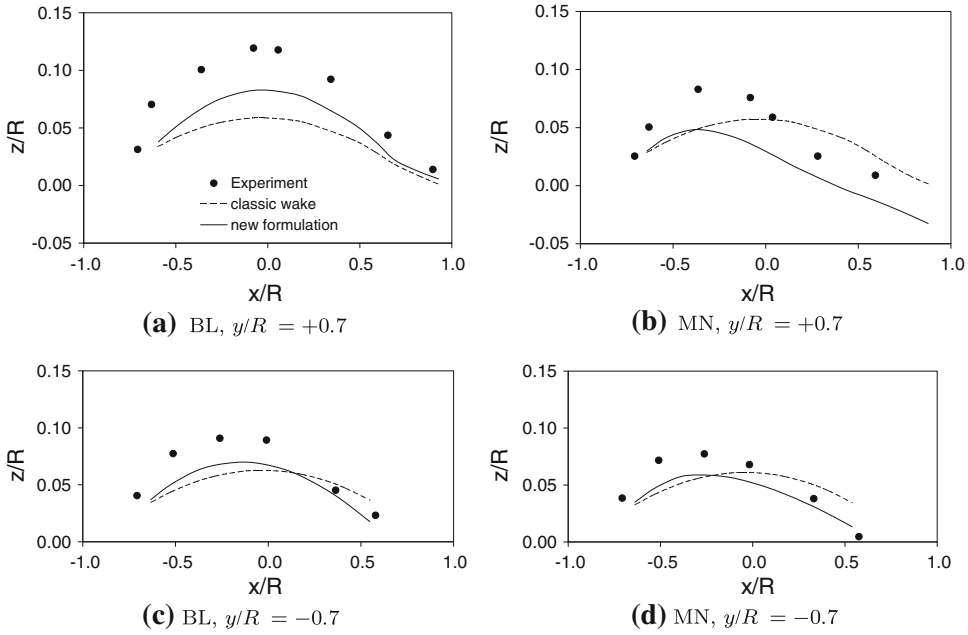


Fig. 15 Wake positions on the advancing and the retreating side of the rotor

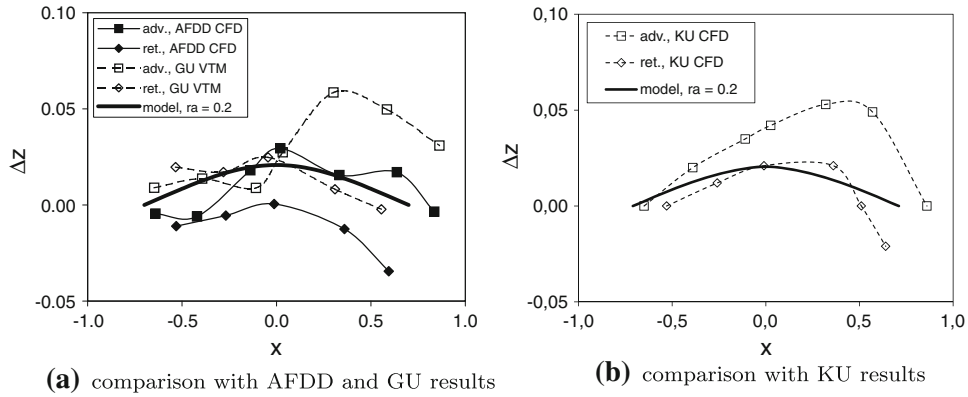


Fig. 16 Effect of the fuselage induced flow on the wake displacement, $|y| = 0.7$

modification of the blade motion as proven by test data [24, 25]. This is one of the major results of the HART I and HART II tests. Exactly this effect is now successfully addressed by the harmonic wake distortion model presented in this paper.

The root cut-out effect can be implemented in a prescribed wake code as has been shown in Sect. 2.3. At least qualitatively this is in good agreement with free-wake results having this effect inherently included.

The inclusion of the fuselage-induced flow field has not yet been validated. No experimental data are available for the HART II data set since the isolated fuselage was not measured. A panel method was applied to provide data that can be used for correlation. In addition, within the International HART II Workshop coupled CFD/CSD computations have been performed, initially without the fuselage, and later with the fuselage [27]. From these data, the difference in wake distortion attributed to the presence of the fuselage can be extracted. In Fig. 16 a comparison is made at $|y| = 0.7$ between the simplified model developed here and results extracted from CFD computations of AFDD [27] and another set of data provided from results obtained with a vorticity transport method of Glasgow University (GU) [28], both shown in Fig. 16a. Since the different computations with and without fuselage were performed with several years between them, some differences in the general set-up and trim conditions may have biased the results. In (b) a comparison is made with recent results of CFD computations performed by Konkuk University (KU), Korea [29]. This latter computation was performed one after the other and is more consistent in the results. The trend of the simplified model developed here is confirmed.

The general trend seen in Fig. 16 is in agreement with the model developed here. The magnitude of the peak induced velocity used for the fuselage model was taken from [20] as $c_1 = 0.15$, (see Eq. 31). In the figure, the

difference in vortex vertical position between isolated rotor and inclusion of the fuselage is shown as $\Delta z = z_v(\text{with fuselage}) - z_v(\text{isolated rotor})$. The advancing (adv.) and retreating (ret.) sides always show a displacement upwards in the center at $x = 0$, relative to the vortex generation at the left side of the figure, or relative to the location where the vortices leave the rotor disk at the right side, indicating the upwash generated by the fuselage before $x = 0$ and the downwash downstream of it.

7 Conclusions

In this paper, several models have been presented that may be applied to any existing basic prescribed wake geometry. These models account for:

1. blade root cut-out in the centre of the hub,
2. harmonic inflow due to harmonic blade loading and the associated wake distortions,
3. stationary inflow due to the presence of the fuselage and the associated wake distortions.

It is found that the root cut-out results in an increased downwash in the remaining part of the disk due to momentum theory. The most visible effect is found behind and downstream of the rotor centre, but it does not affect the locations of blade-vortex interaction in the first and fourth quadrants.

A major and very important contribution results from the new harmonic inflow model and the wake distortion caused by it. This is a key parameter for a reliable computation of BVI locations in the rotor disk, and the noise radiation generated. The application of only the fundamental basic wake geometry cannot predict the BVI locations correctly since it does not modify the wake geometry.

The inclusion of the fuselage-induced flow field and the associated wake distortion is mainly found along the centreline of the rotor disk and to the right and left of it, while it quickly reduces when moving more far away from the fuselage in the lateral direction. Thus, its effect on BVI location is not large, but further validation remains to be done. It represents essential physics in a plausible manner and should be taken into account in future.

All of these models are able to be superimposed to each other and to any available prescribed basic wake model. This allows them to be added without modification of the existing applied wake. All wake distortions due to harmonic loading and fuselage directly are additive. This modularity appears as a major advantage.

In total, with the addition of these models the prescribed wake codes are significantly improved and their applicability enhanced. The remaining gap in accuracy compared to free-wake codes is recouped by the significant amount of computation time saved when using prescribed wake codes compared to free-wake codes.

Acknowledgments The provision of data from CFD/CSD computations (Joon W. Lim, US Army AFDD) and from the Vorticity Transport Method (VTM; R. Brown and M. Kelly, University of Glasgow) in Fig. 16 is highly appreciated. The CFD data of the fuselage flow in Fig. 10c, d as well as those of Fig. 16b are kindly provided by S. Jung of Konkuk University in Seoul, Korea.

References

1. International Standards and Recommended Practices: Aircraft Noise, ICAO, Annex 16, vol. 1, USA (1988)
2. Splettstößer, W.R., Schultz, K.-J., Kube, R., Brooks, T.F., Booth, E.R. Jr., Niesl, G., Streby, O.: A higher harmonic control test in the DNW to reduce impulsive BVI noise. J. Am. Helicopter Soc. **39**(4), 3–13 (1994)
3. McCroskey, W.J.: Vortex wakes of rotorcraft. In: 33rd Aerospace Sciences Meeting and Exhibit, Reno, NV, USA, January (1995)
4. Gray, R.B.: An aerodynamic analysis of a single-bladed rotor in hovering and low speed forward flight as determined from the smoke studies of the vorticity distribution in the wake. Princeton Aeronautical Engineering Department Report 356 (1956)
5. Landgrebe, A.J.: An analytical and experimental investigation of helicopter rotor hovering performance and wake geometry characteristics, United States Army Air Mobility Research and Development Laboratory, USAAMRDL TR 71-24 (1971)
6. Beddoes, T.S.: A Wake model for high resolution airloads. In: International Conference on Helicopter Basic Research, Triangle Park, NC, February (1985)
7. Scully, M.P.: A method of computing helicopter vortex wake distortion, Massachusetts Institute of Technology, Report ASRL TR 138-1 (1967)
8. Bhagwat, M.J., Leishman, J.G.: Stability, consistency and convergence of time-marching free-vortex rotor wake algorithms. J. Am. Helicopter Soc. **46**(1), 59–71 (2001)
9. Lim, J.W.: Computational modeling of HART II blade–vortex interaction loading and wake system. J. Aircr. **45**(3), 923–933 (2008)
10. Rankine, W.J.M.: On the mechanical principles of the action of propellers. Trans. Inst. Naval Archit. **VI**, 13–30 (1865)
11. Froude, W.: On the elementary relation between pitch, slip and propulsive efficiency. Trans. Inst. Naval Archit. **XIX**, 47–57 (1878)
12. Glauert, H.: A general theory of the Autogyro. Aeronautical Reseach Council, Reports and Memoranda 1111 (1926)
13. Lock, C.N.H.: Further development of the Autogyro Theory. Aeronautical Reseach Council, Reports and Memoranda 1127 (1927)
14. Drees, J.M.: A theory of airflow through rotors and its application to some helicopter problems. J. Helicopter Assoc. G. B. **3**(2), 79–104 (1949)
15. Johnson, W.: Helicopter Theory. Princeton University Press, Princeton (1980)
16. Leishman, J.G.: Principles of helicopter aerodynamics. Cambridge University Press, New York (2000)
17. Mangler, K.W., Squire, H.B.: The induced velocity field of a rotor. Aeronautical Reseach Council, Reports and Memoranda 2642 (1950)
18. van der Wall, B.G.: The effect of HHC on the vortex convection in the wake of a helicopter rotor. Aerospace Science and Technology, vol. 4 (5), July (2000)

19. Bronstein, I.N., Semendjajew, K.A.: Handbook of Mathematics, 5th edn. Springer, Berlin (2007)
20. van der Wall, B.G., Göpel, C.: Über den Einfluß der Rotorversuchsstände ROTEST und ROTOS auf die Rotordurchströmung im DNW, DLR-Mitteilungen 91-16, September (1991)
21. van der Wall, B.G., Yin, J.: DLR's S4 Rotor Code Validation With HART II Data: the baseline case. In: 1st International Forum on Rotorcraft Multidisciplinary Technology, Seoul, Korea, October (2007)
22. van der Wall, B.G., Yin, J.: Simulation of active rotor control by comprehensive rotor code with prescribed wake using HART II Data, 65th AHS Forum, Grapevine, TX, USA, May (2009)
23. van der Wall, B.G., Yin, J.: Cruise-speed BVI noise computation by comprehensive code and validation with HART I Data. In: 2nd International Forum on Rotorcraft Multidisciplinary Technology, Seoul, Korea, October (2009)
24. van der Wall, B.G., Burley, C.L., Yu, Y.H., Pengel, K., Beaumier, P.: The HART II test—measurement of helicopter rotor wakes. Aerosp. Sci. Technol. **8**(4), 273–284 (2004)
25. Kube, R., Splettstößer, W.R., Wagner, W., Seelhorst, U., Yu, Y.H., Boutier, A., Mercker, E.: Initial results from the higher harmonic aeroacoustic rotor test (HART) in the German–Dutch Wind Tunnel. In: 75th AGARD Fluid Dynamics Panel Meeting and Symposium on Aerodynamics and Acoustics of Rotorcraft, Berlin, Germany, October (1994)
26. Yin, J., Delfs, J.: Improvement of the DLR rotor aeroacoustic Code (APSIM) and its validation with analytic solution. In: 29th European Rotorcraft Forum, Friedrichshafen, Germany, September (2003)
27. Lim, J.W.: The effect of fuselage and rotor hub on blade–vortex interaction airloads and rotor wakes. In: 36th European Rotorcraft Forum, Paris, France, September (2010)
28. Kelly, M.E., Brown, R.: Predicting the wake structure of the HART II rotor using the vorticity transport model. In: 34th European Rotorcraft Forum, Liverpool, UK, September (2008)
29. Park, J.-S., Sa, J.-H., You, Y.-H., Jung, S.N., Park, S.-H.: Fuselage effect in multibody dynamics/CFD coupled analysis for a rotor correlation in descending flight. In: 1st Asian Australian Rotorcraft Forum and Exhibition, Busan, Korea, February (2012)